

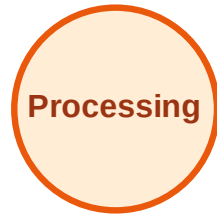
# BFS Traversal

Step 1: Start at A (Seeking target node K)

## Legend



Complete



Processing



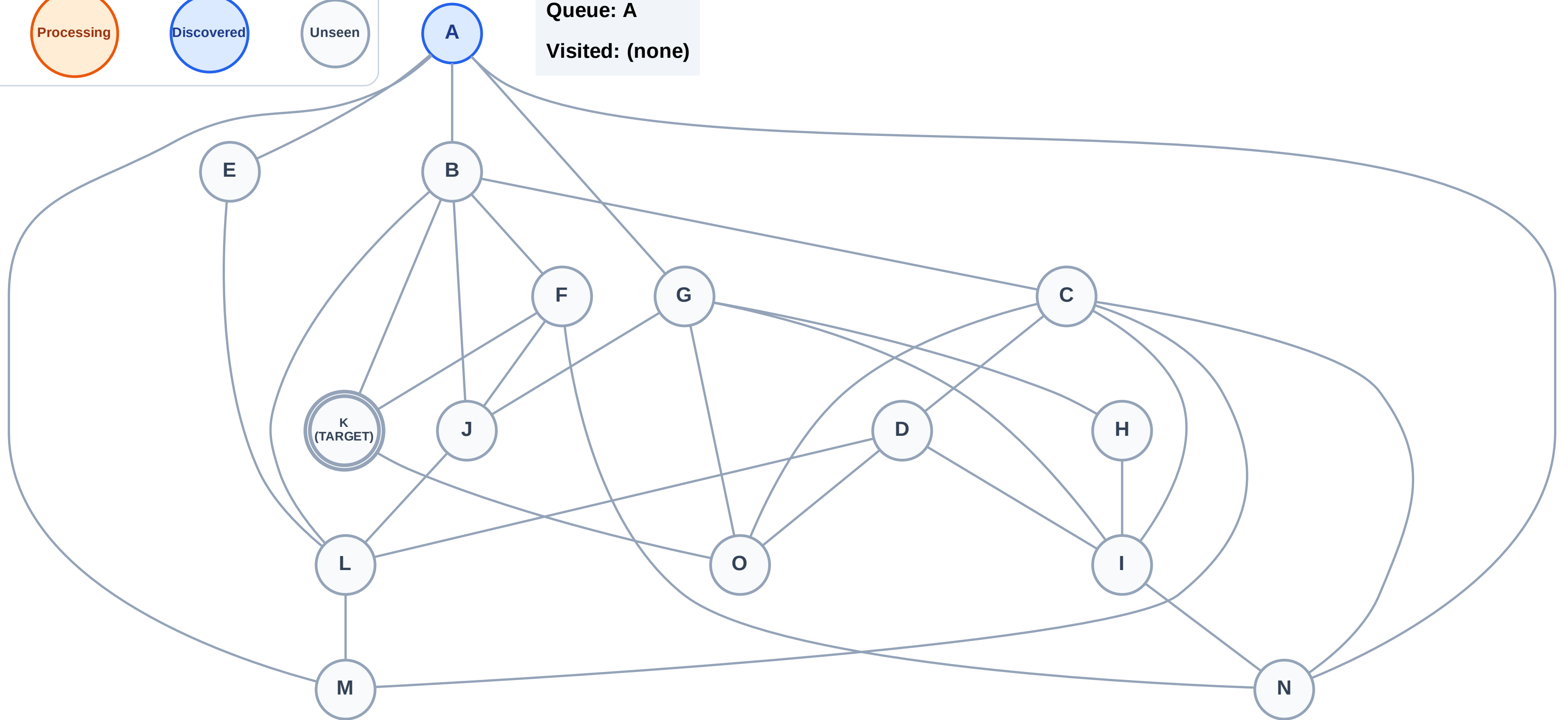
Discovered



Unseen

Queue: A

Visited: (none)



# BFS Traversal

Step 2: Process node A

Legend

Complete

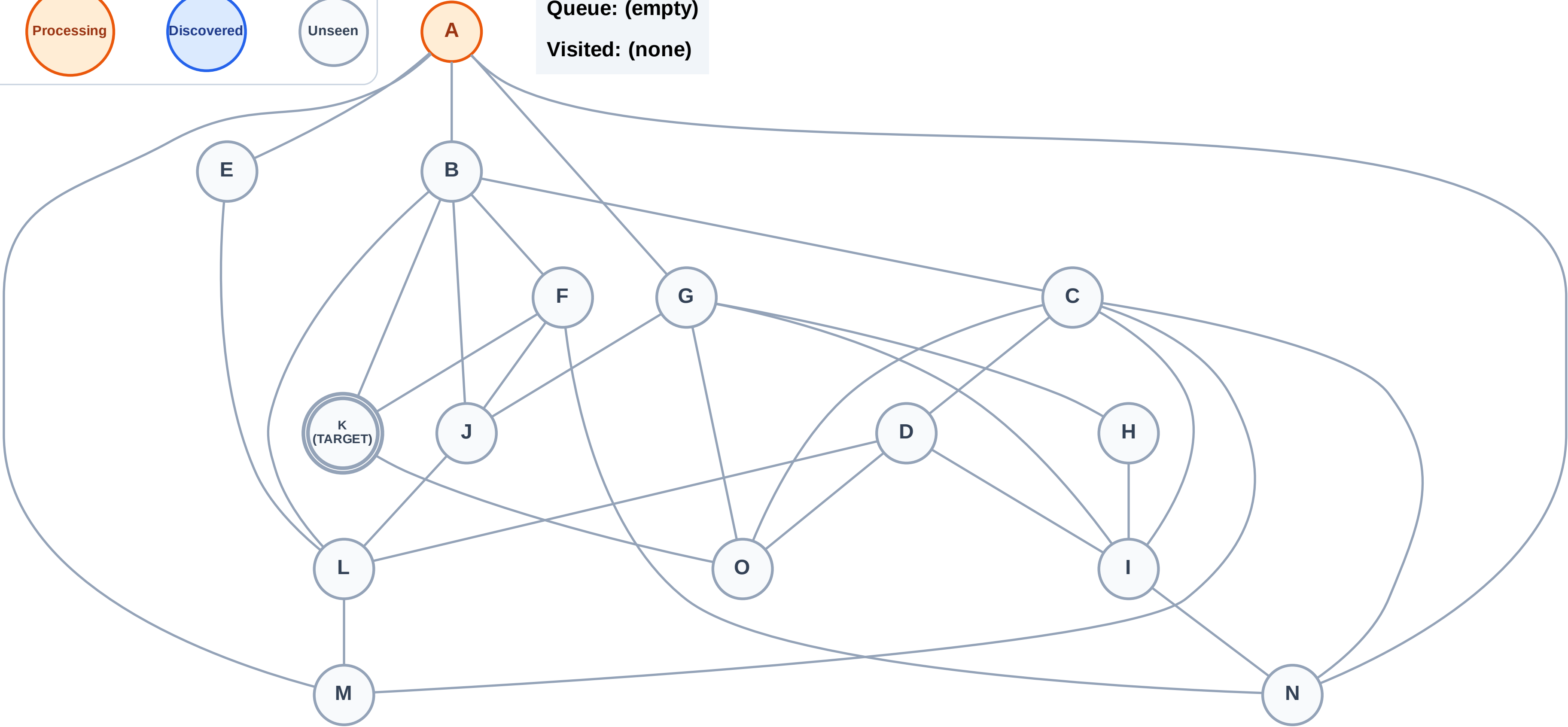
Processing

Discovered

Unseen

Queue: (empty)

Visited: (none)



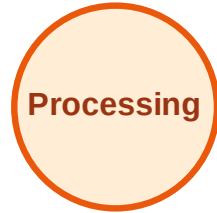
# BFS Traversal

Step 3: Discover B, E, G, M, N from A

## Legend



Complete



Processing



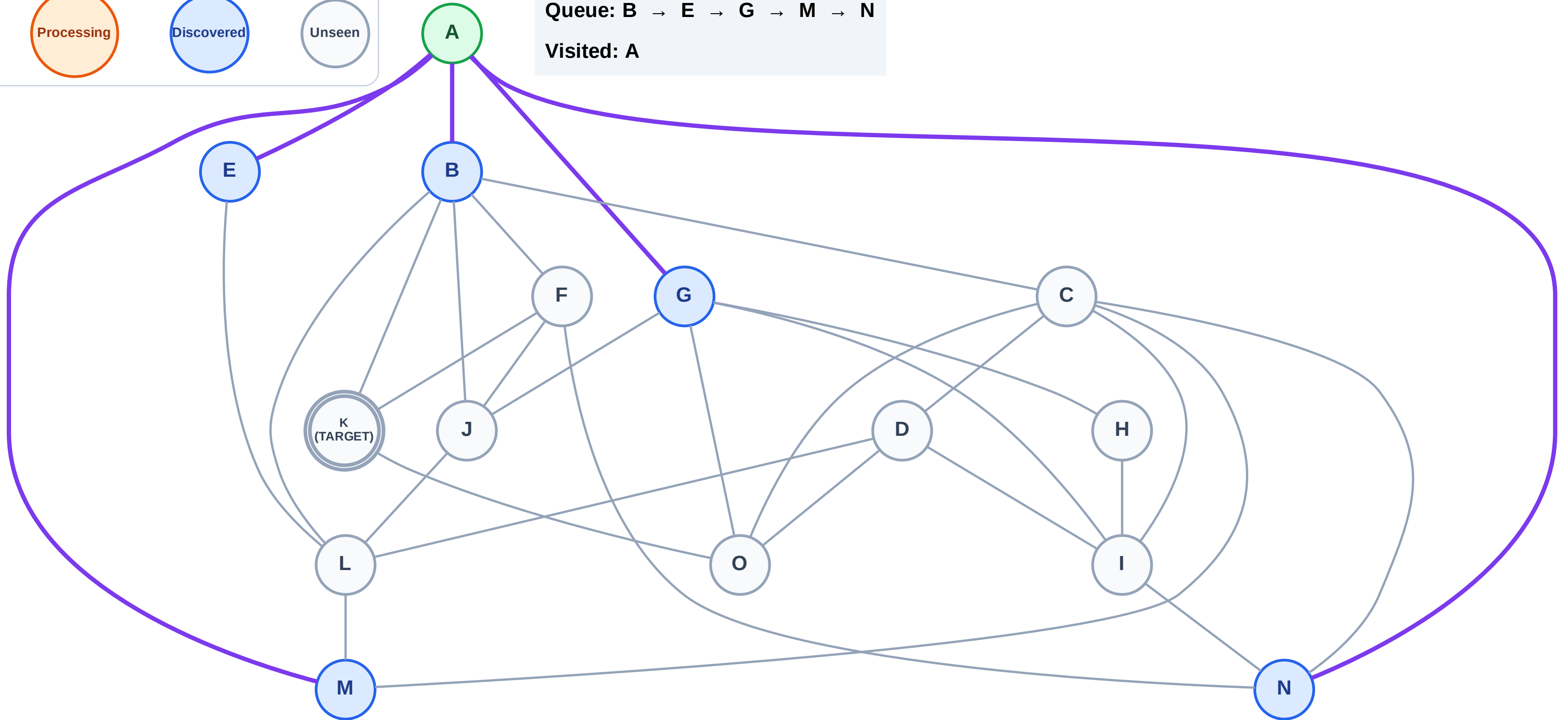
Discovered



Unseen

Queue: B → E → G → M → N

Visited: A



# BFS Traversal

Step 4: Process node B

Legend

Complete

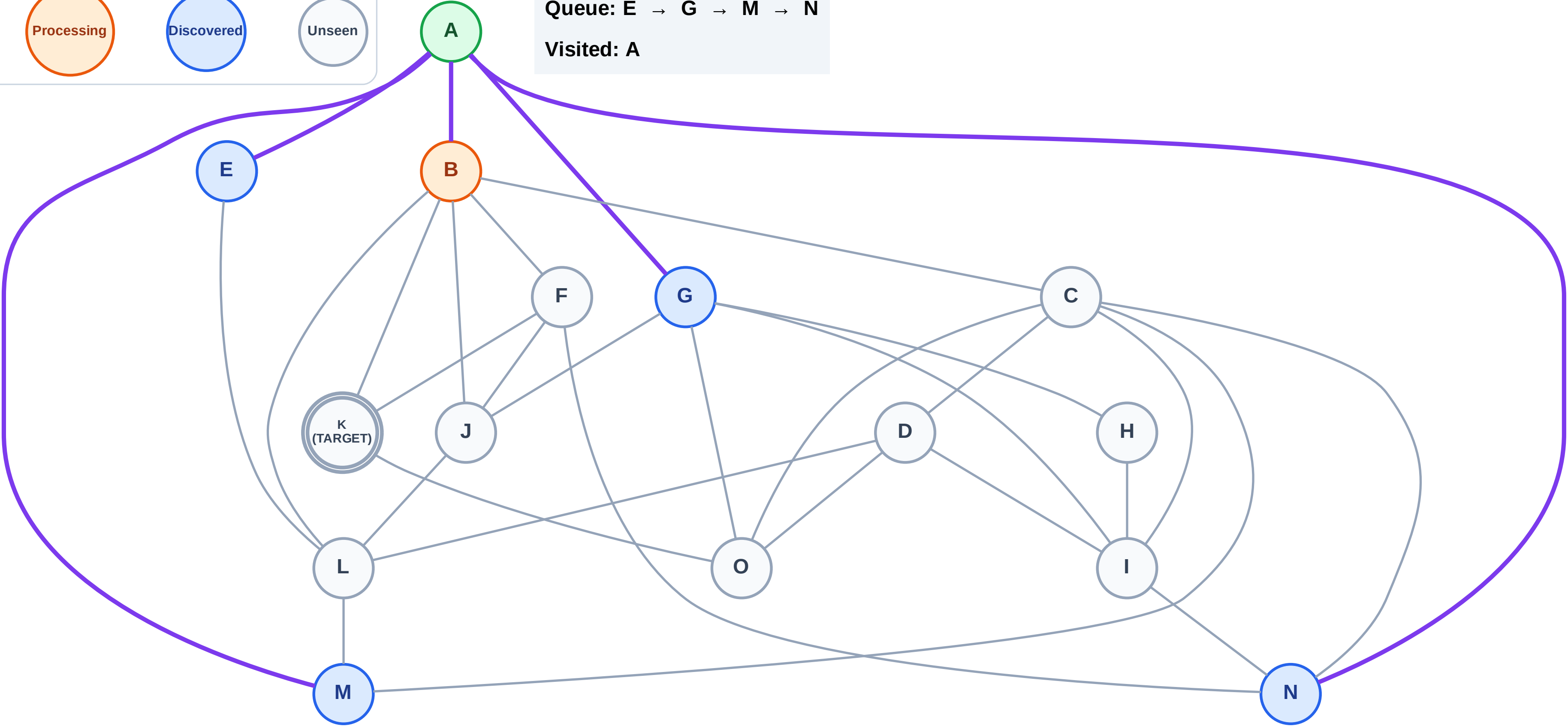
Processing

Discovered

Unseen

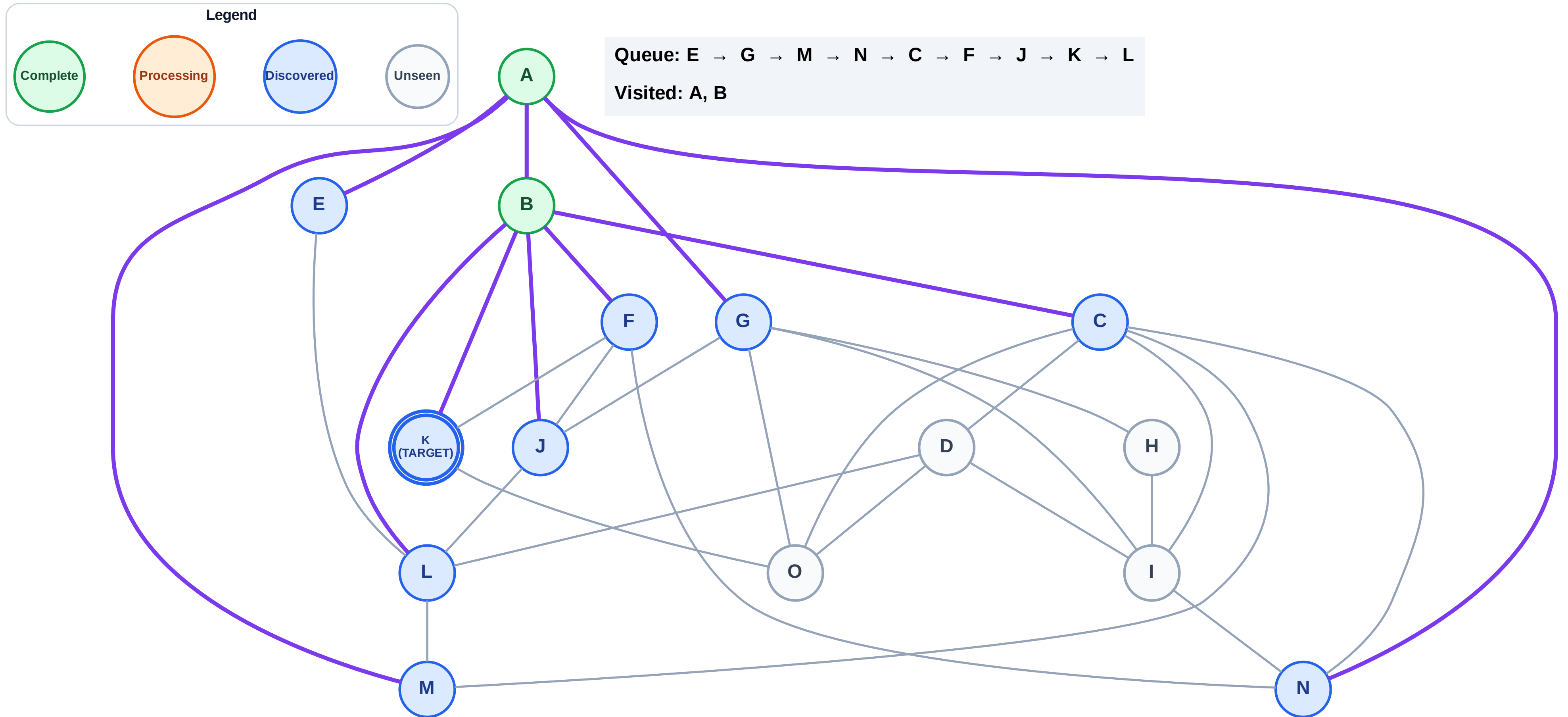
Queue: E → G → M → N

Visited: A



### Step 5: Discover C, F, J, K, L from B

### Step 5: Discover C, F, J, K, L from B



# BFS Traversal

Step 6: Process node E

Legend

Complete

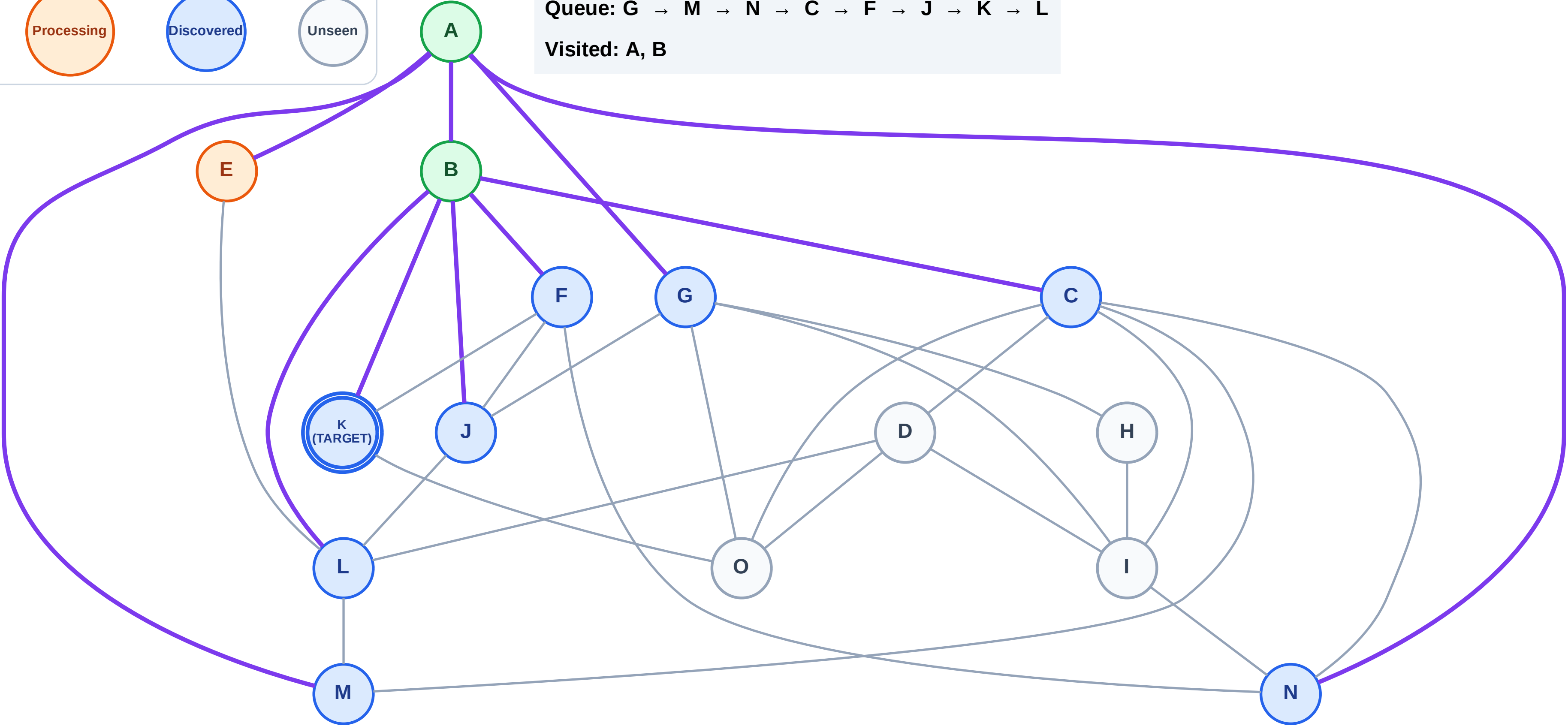
Processing

Discovered

Unseen

Queue: G → M → N → C → F → J → K → L

Visited: A, B



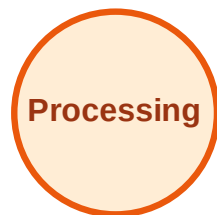
# BFS Traversal

Step 7: No new neighbors from E

## Legend



Complete



Processing



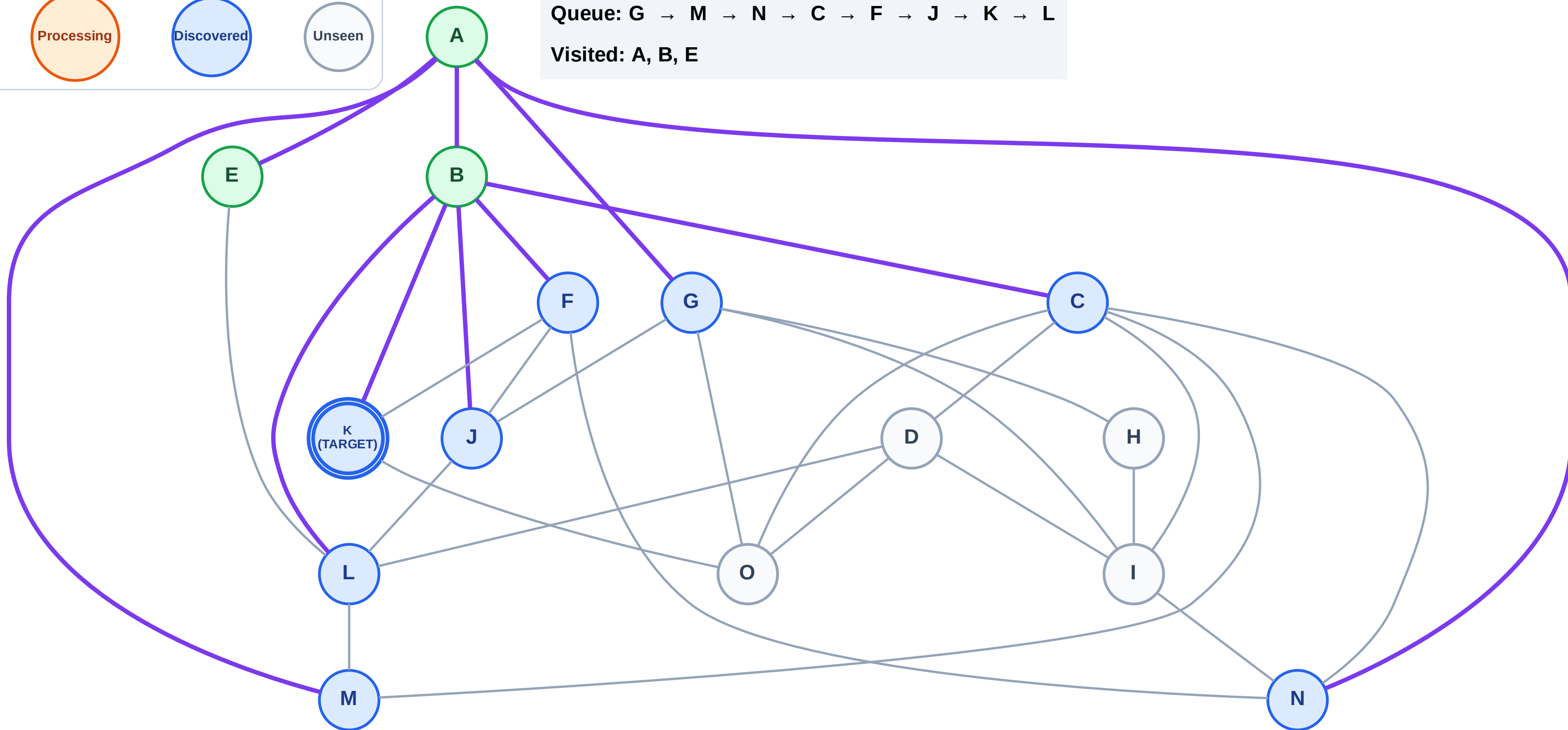
Discovered



Unseen

Queue: G → M → N → C → F → J → K → L

Visited: A, B, E



# BFS Traversal

Step 8: Process node G

Legend

Complete

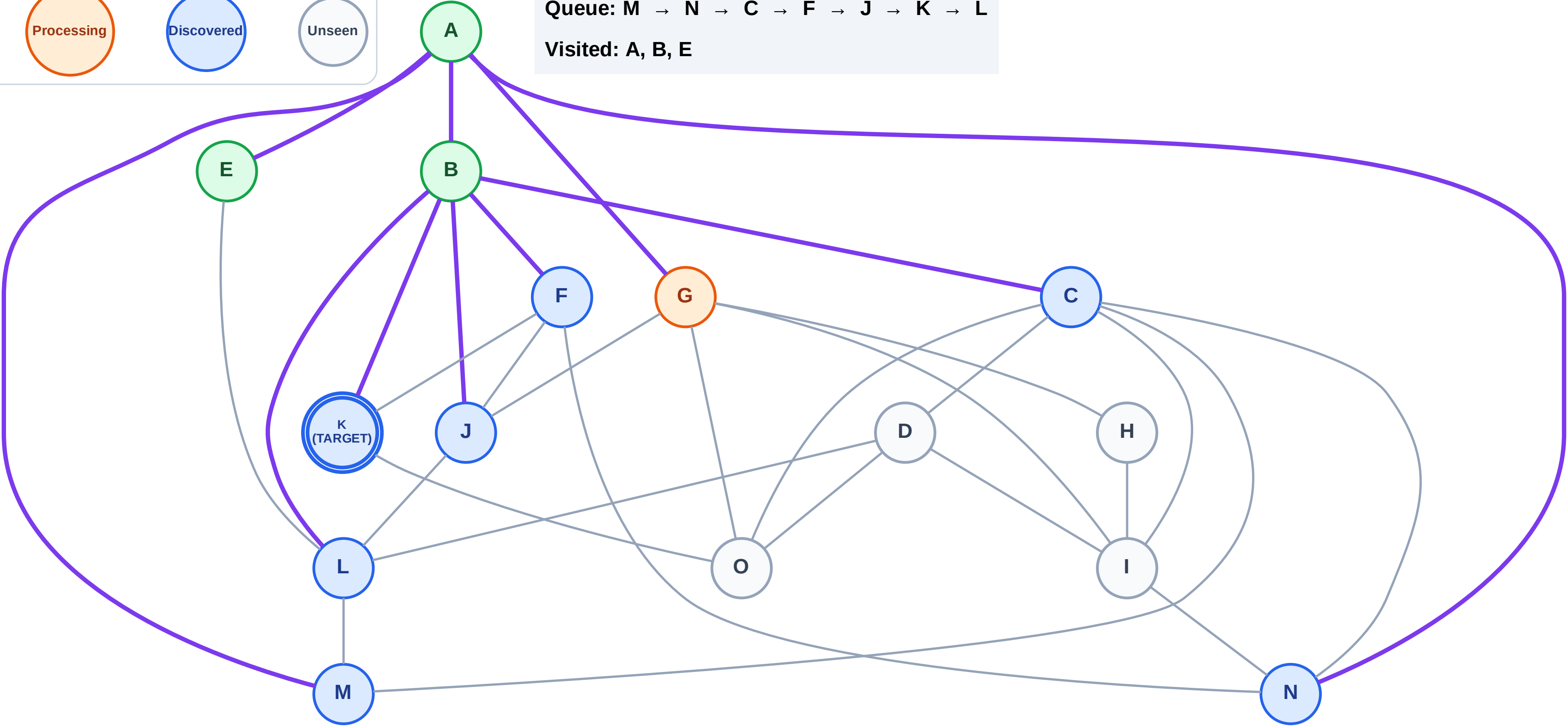
Processing

Discovered

Unseen

Queue: M → N → C → F → J → K → L

Visited: A, B, E



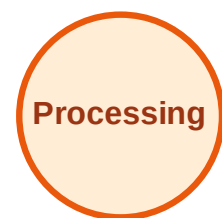
# BFS Traversal

Step 9: Discover H, I, O from G

## Legend



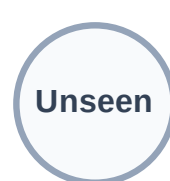
Complete



Processing



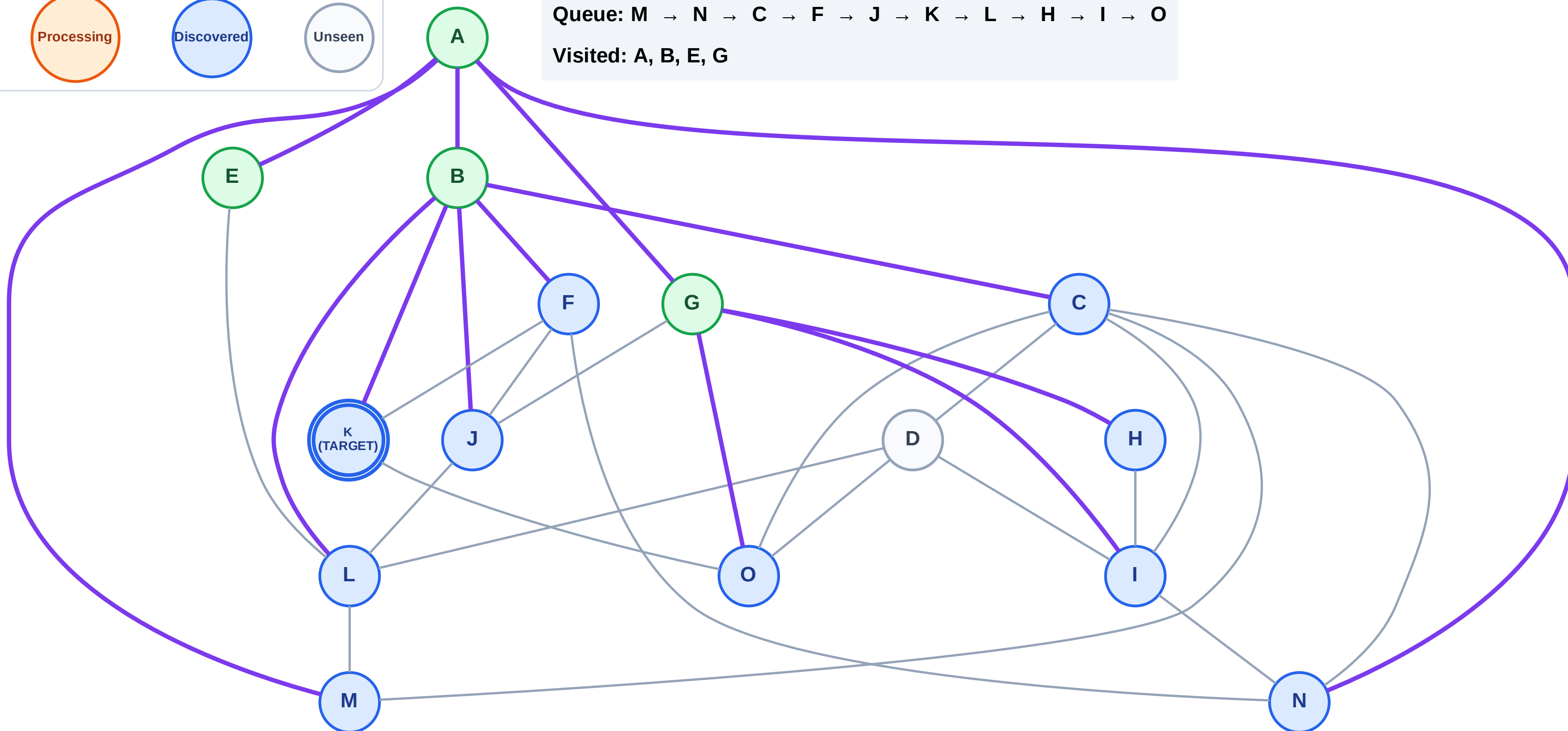
Discovered



Unseen

Queue: M → N → C → F → J → K → L → H → I → O

Visited: A, B, E, G



# BFS Traversal

Step 10: Process node M

Legend

Complete

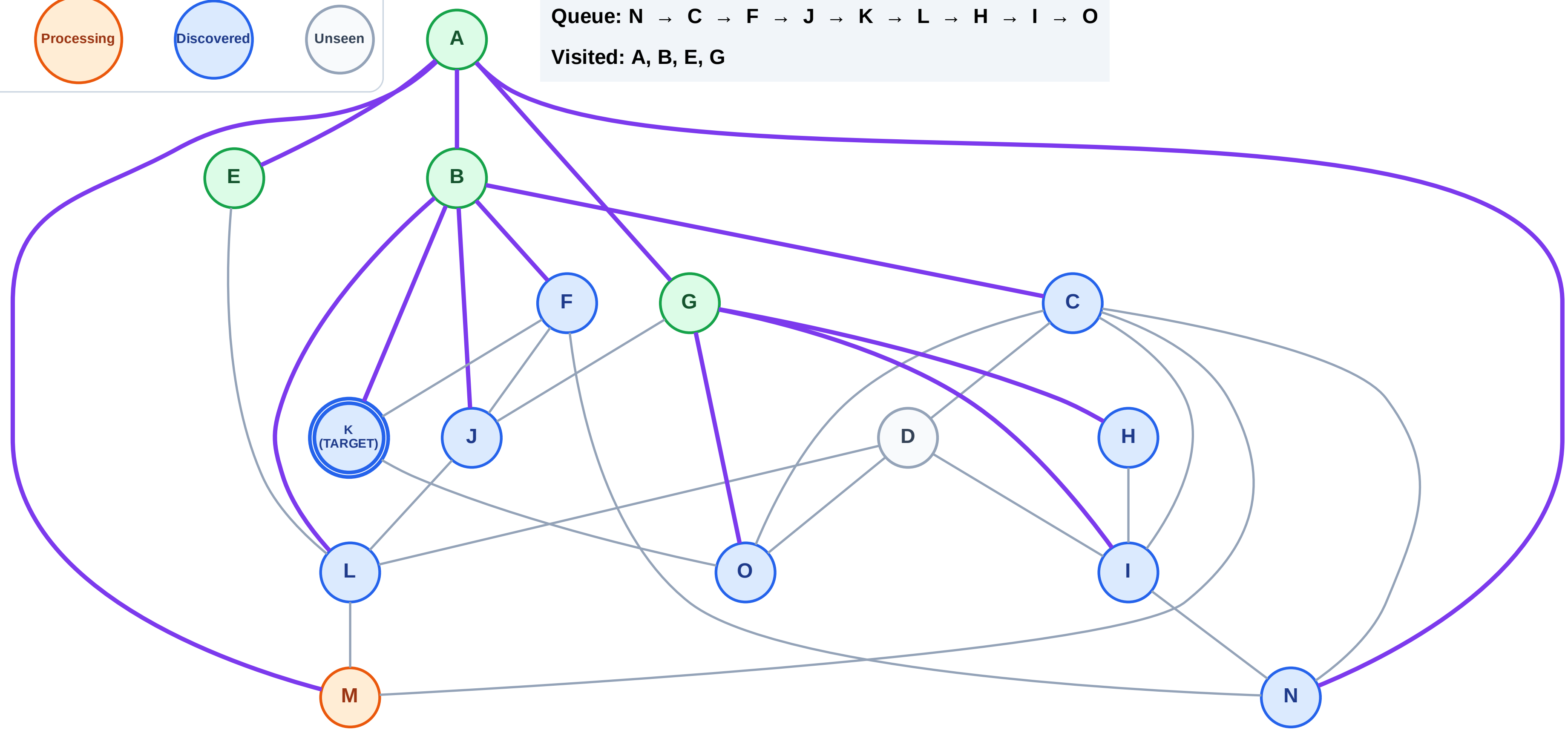
Processing

Discovered

Unseen

Queue: N → C → F → J → K → L → H → I → O

Visited: A, B, E, G



# BFS Traversal

Step 11: No new neighbors from M

## Legend



Complete



Processing



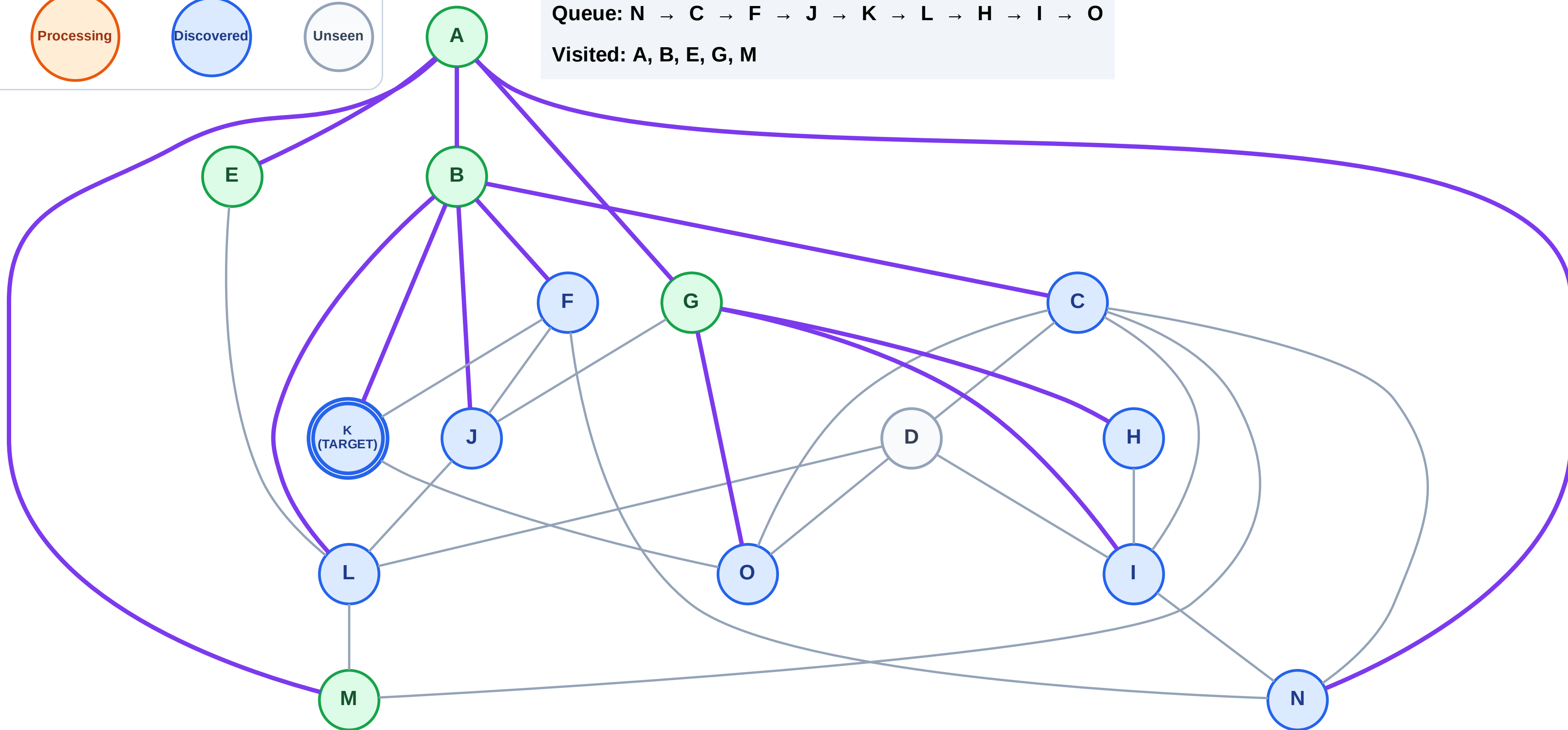
Discovered



Unseen

Queue: N → C → F → J → K → L → H → I → O

Visited: A, B, E, G, M



# BFS Traversal

Step 12: Process node N

Legend

Complete

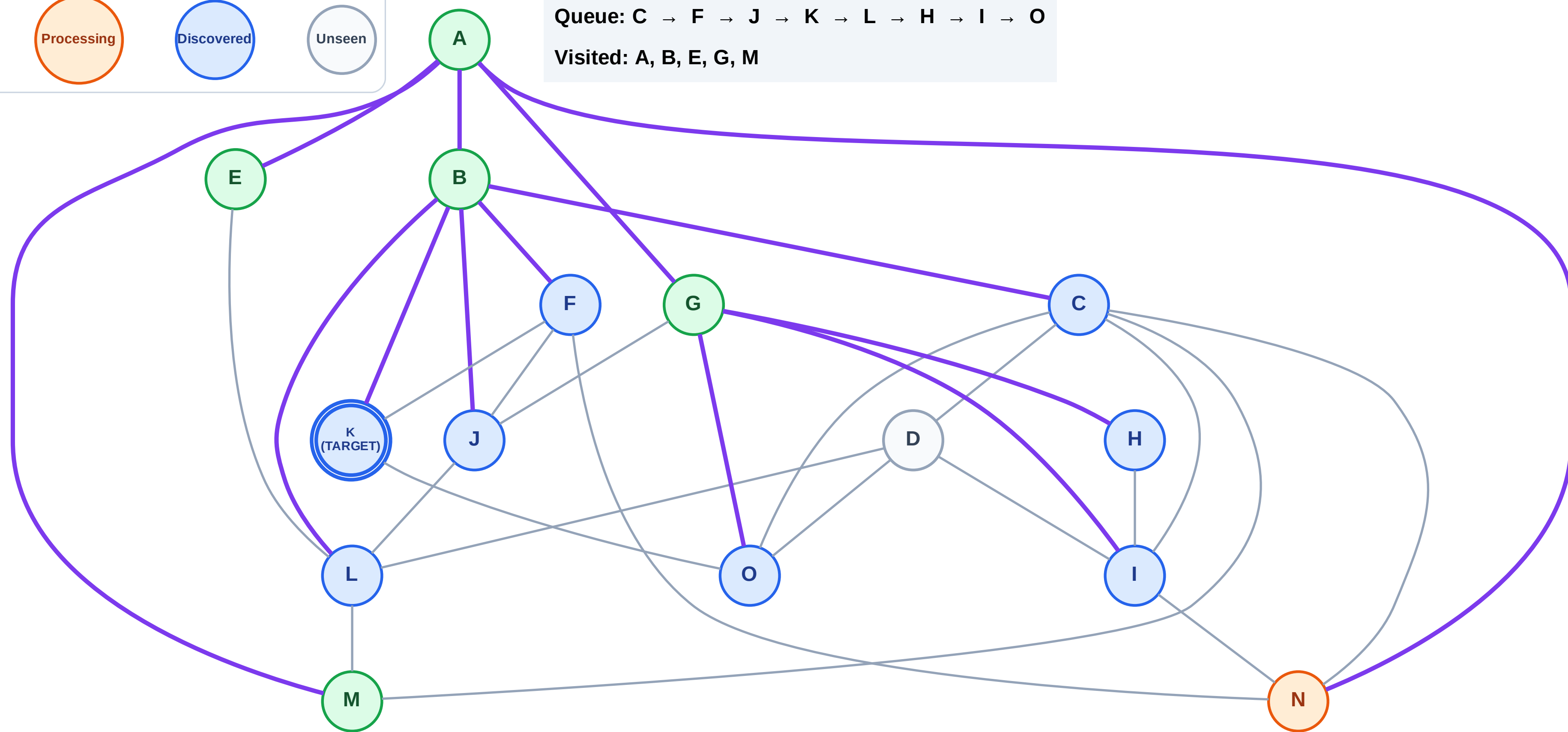
Processing

Discovered

Unseen

Queue: C → F → J → K → L → H → I → O

Visited: A, B, E, G, M



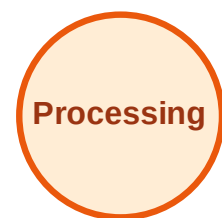
# BFS Traversal

Step 13: No new neighbors from N

## Legend



Complete



Processing



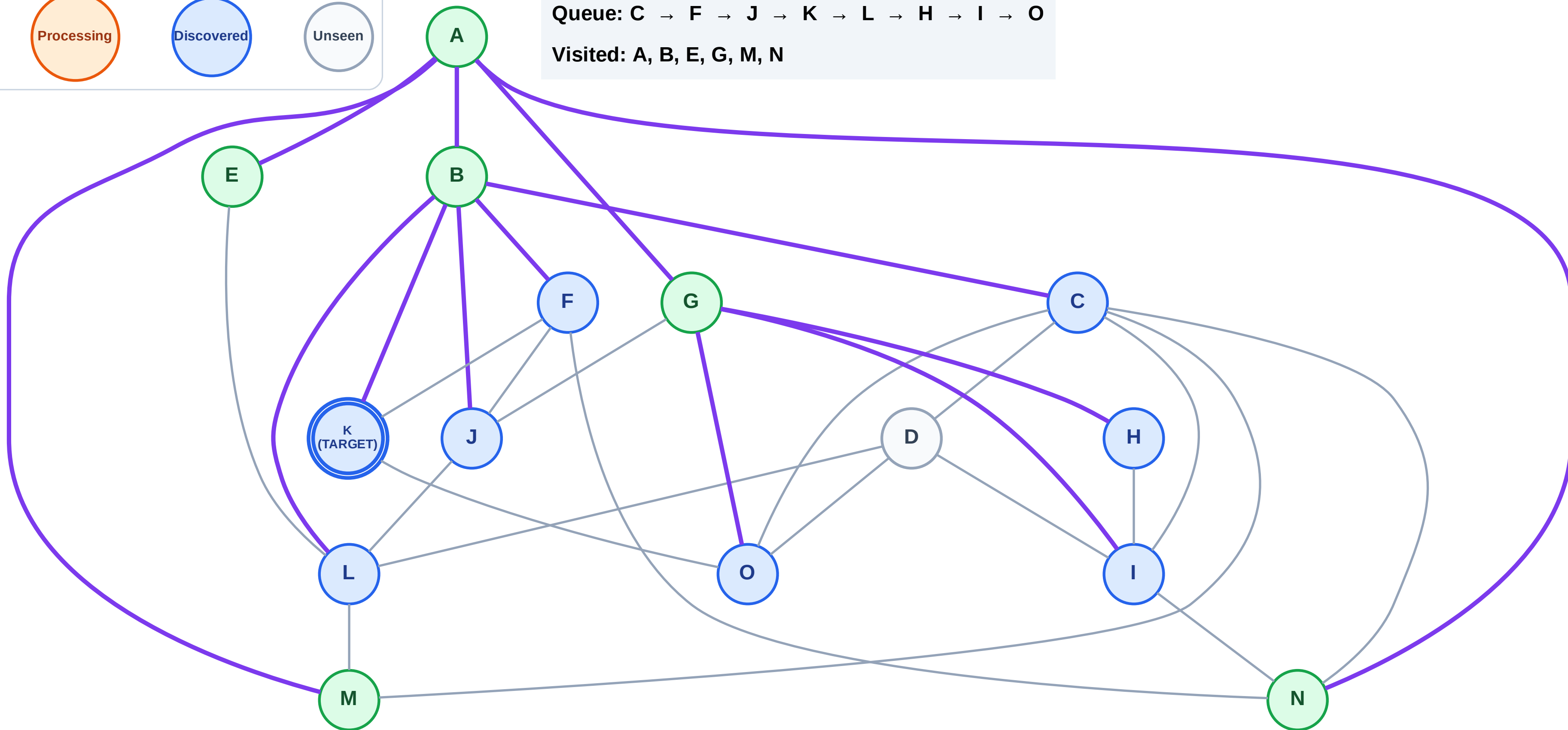
Discovered



Unseen

Queue: C → F → J → K → L → H → I → O

Visited: A, B, E, G, M, N



# BFS Traversal

Step 14: Process node C

Legend

Complete

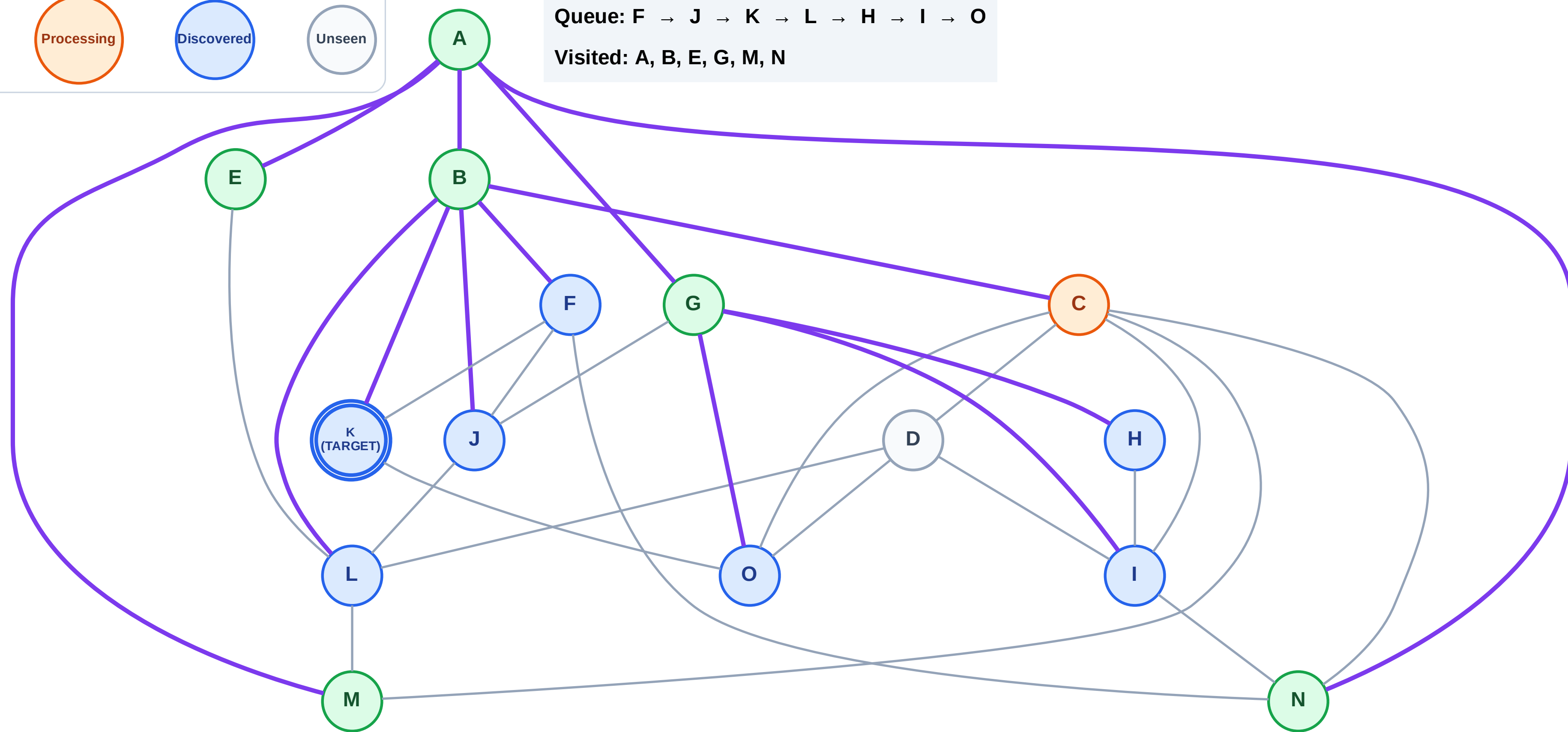
Processing

Discovered

Unseen

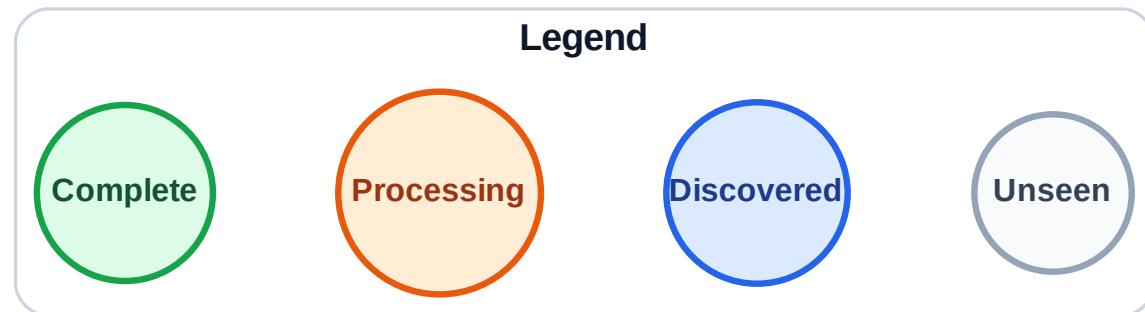
Queue: F → J → K → L → H → I → O

Visited: A, B, E, G, M, N



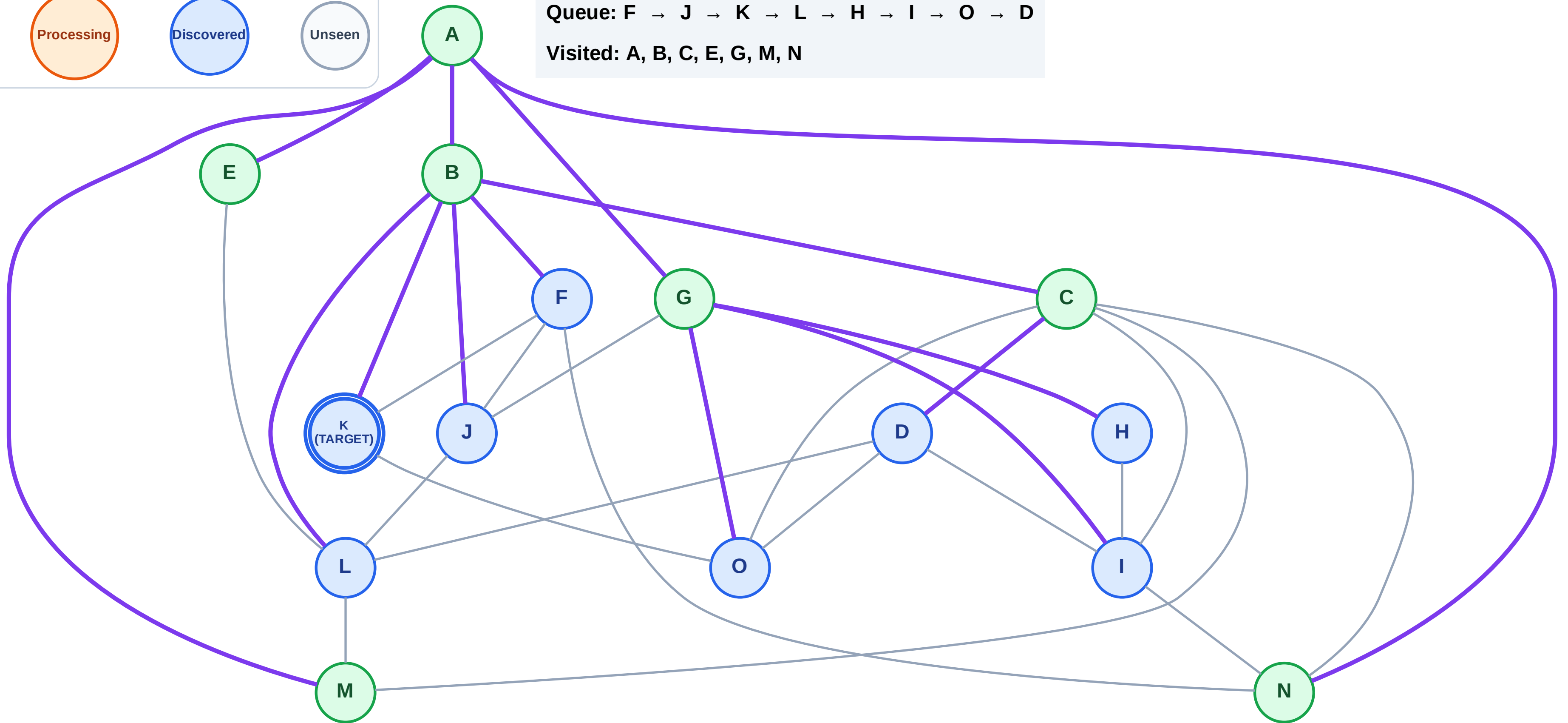
# BFS Traversal

Step 15: Discover D from C



Queue: F → J → K → L → H → I → O → D

Visited: A, B, C, E, G, M, N



# BFS Traversal

Step 16: Process node F

Legend

Complete

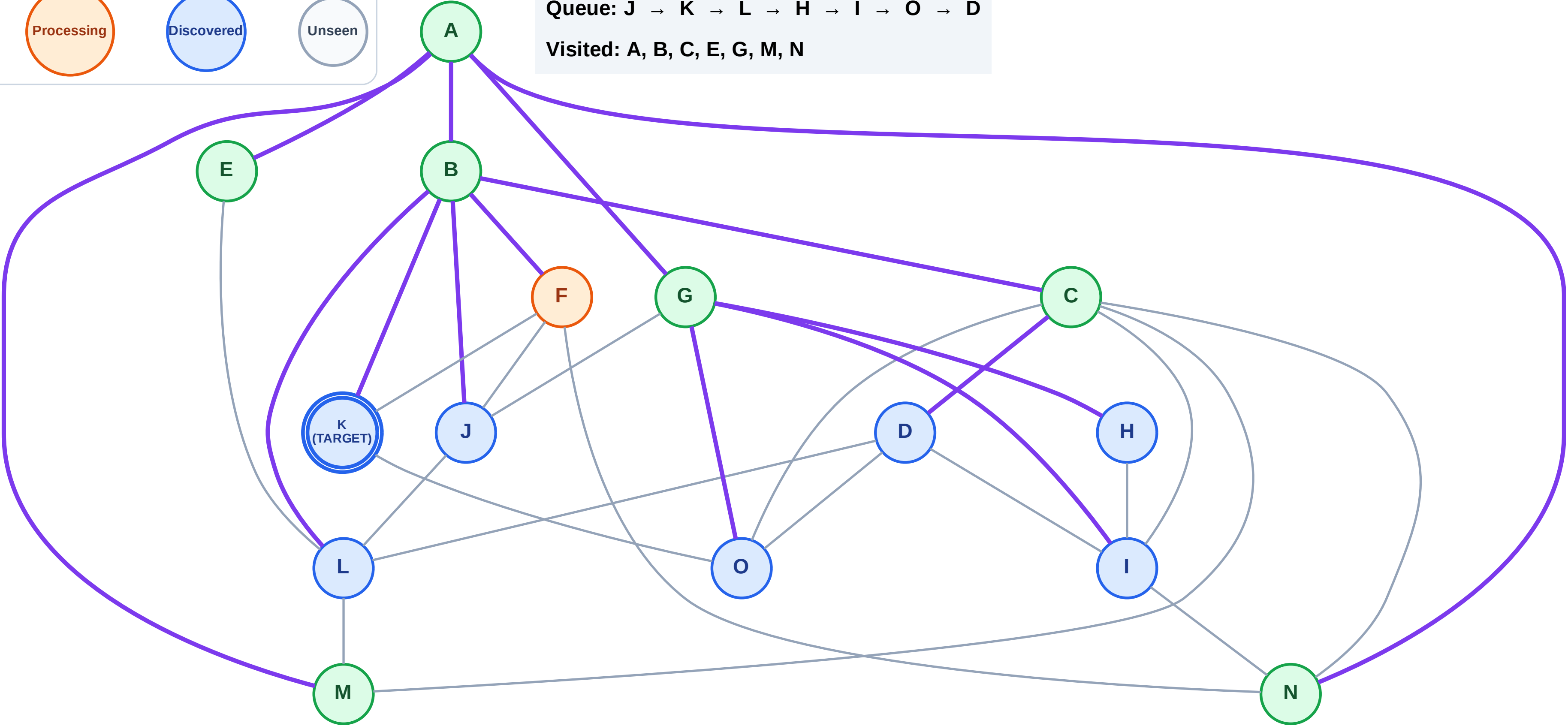
Processing

Discovered

Unseen

Queue: J → K → L → H → I → O → D

Visited: A, B, C, E, G, M, N



# BFS Traversal

Step 17: No new neighbors from F

## Legend



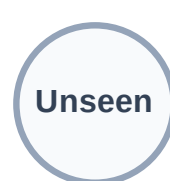
Complete



Processing



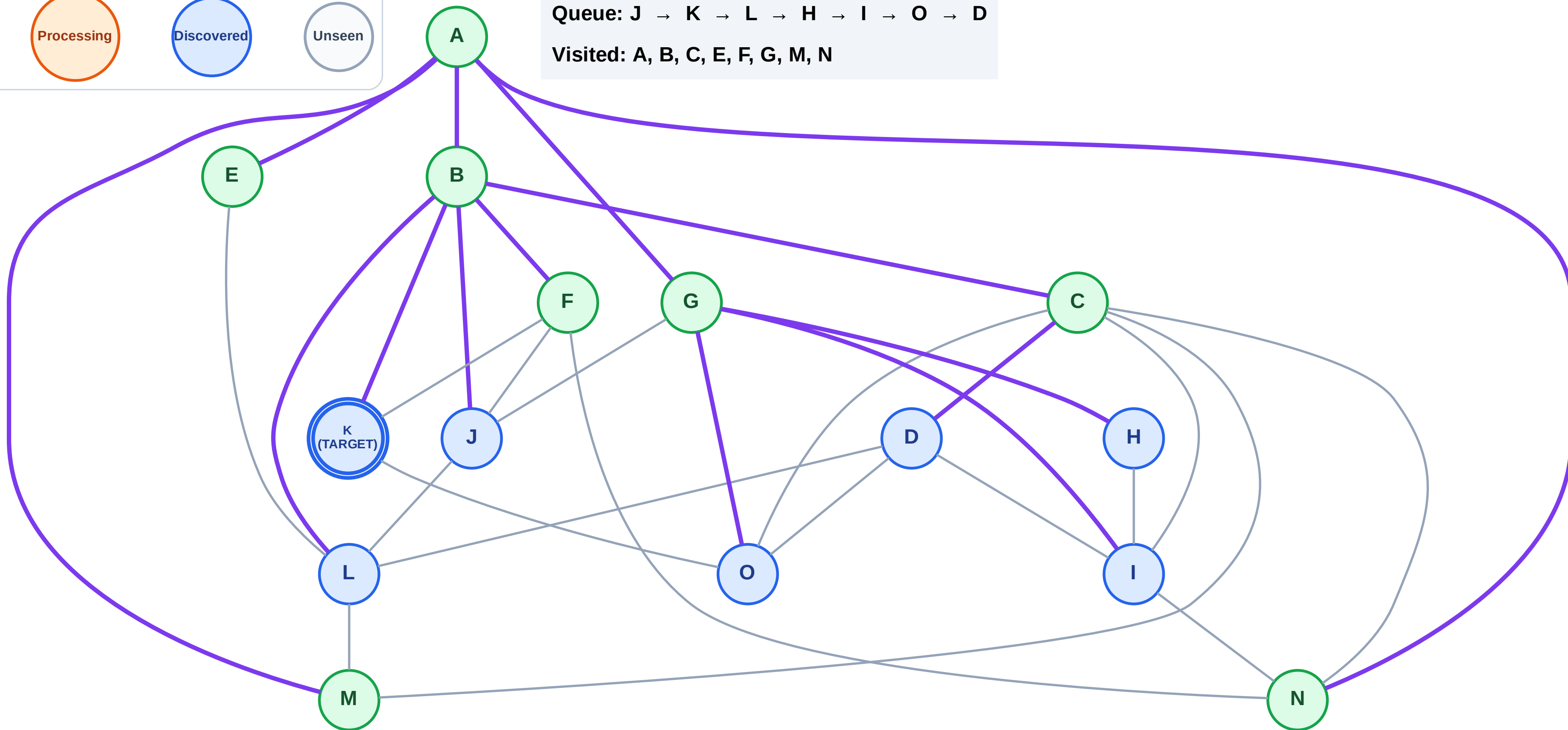
Discovered



Unseen

Queue: J → K → L → H → I → O → D

Visited: A, B, C, E, F, G, M, N



# BFS Traversal

Step 18: Process node J

Legend

Complete

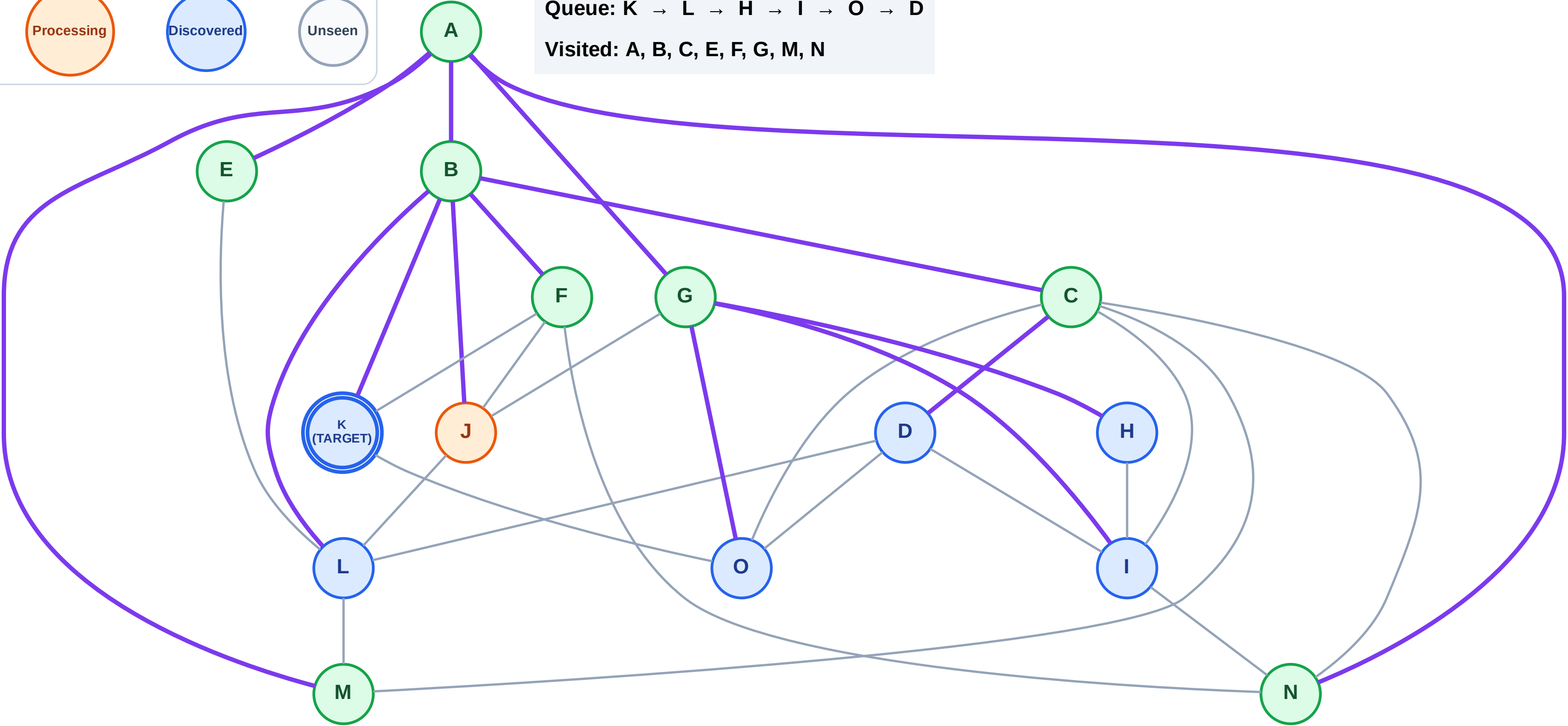
Processing

Discovered

Unseen

Queue: K → L → H → I → O → D

Visited: A, B, C, E, F, G, M, N



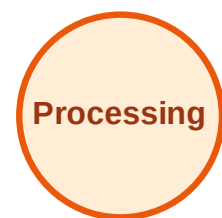
# BFS Traversal

Step 19: No new neighbors from J

## Legend



Complete



Processing



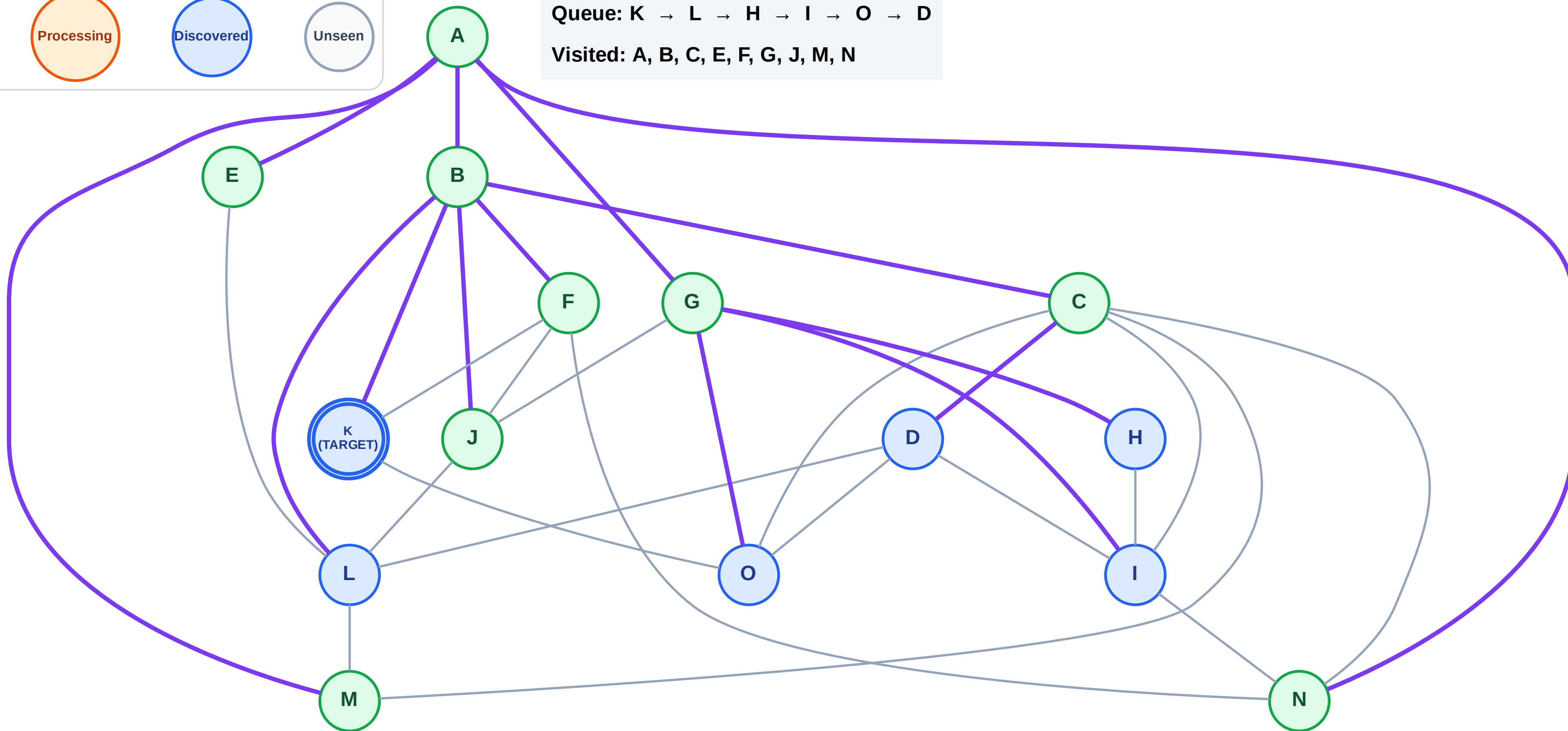
Discovered



Unseen

Queue: K → L → H → I → O → D

Visited: A, B, C, E, F, G, J, M, N



# BFS Traversal

Step 20: Process node K

Legend

Complete

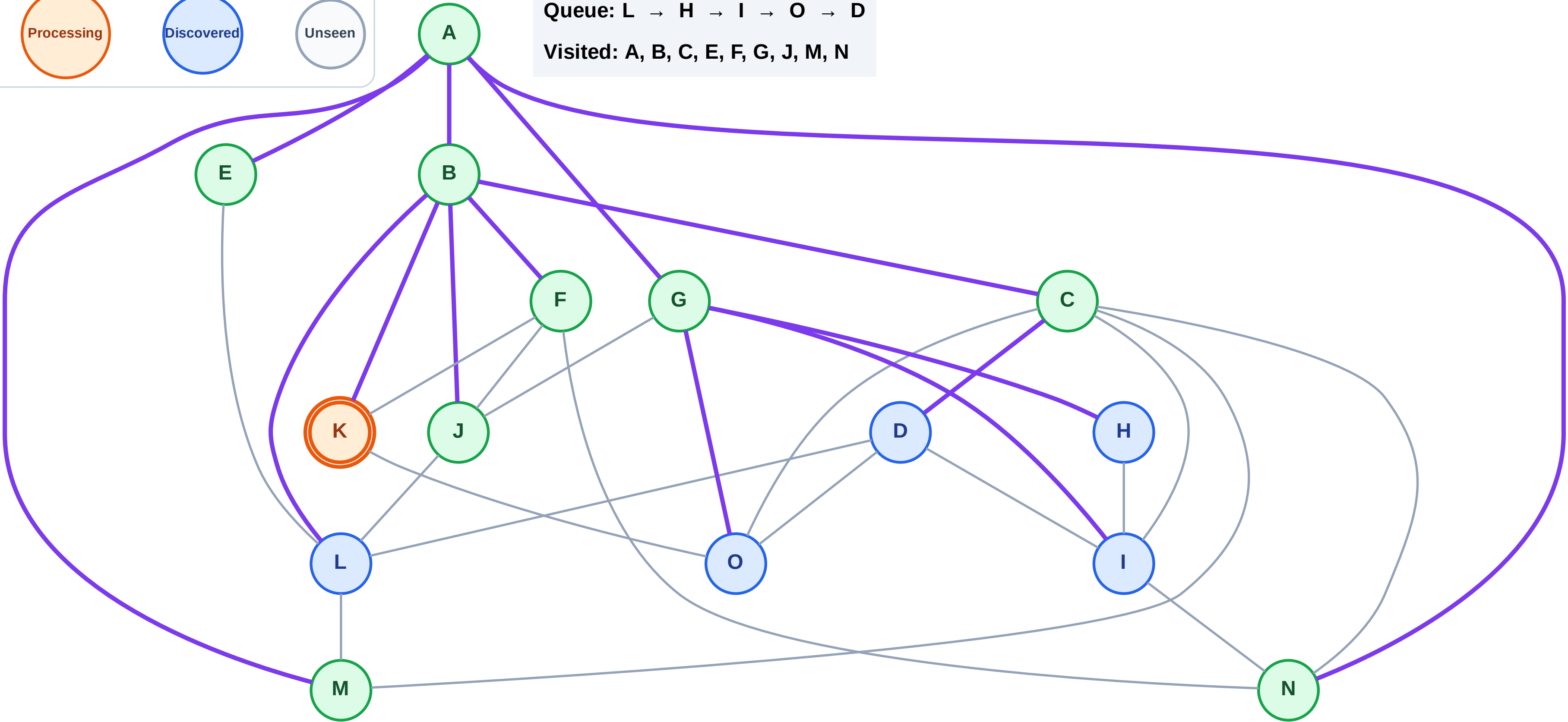
Processing

Discovered

Unseen

Queue: L → H → I → O → D

Visited: A, B, C, E, F, G, J, M, N



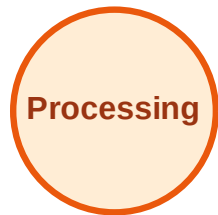
# BFS Traversal

Step 21: TARGET NODE K REACHED! Path (2 hops): A → B → K

## Legend



Complete



Processing



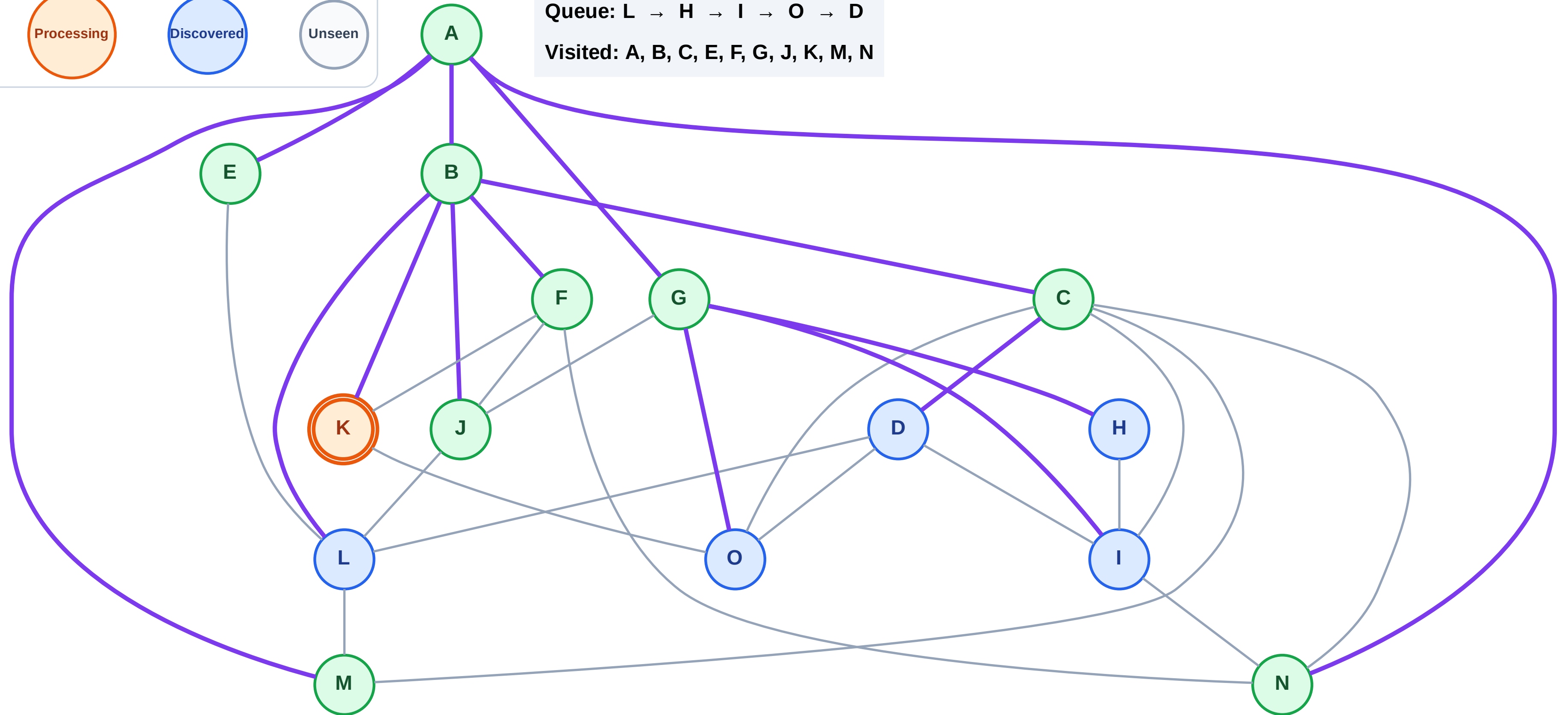
Discovered



Unseen

Queue: L → H → I → O → D

Visited: A, B, C, E, F, G, J, K, M, N



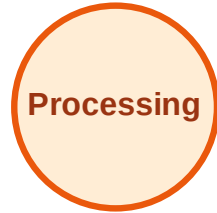
# DFS Traversal

Step 1: Start at A (Seeking target node K)

## Legend



Complete



Processing



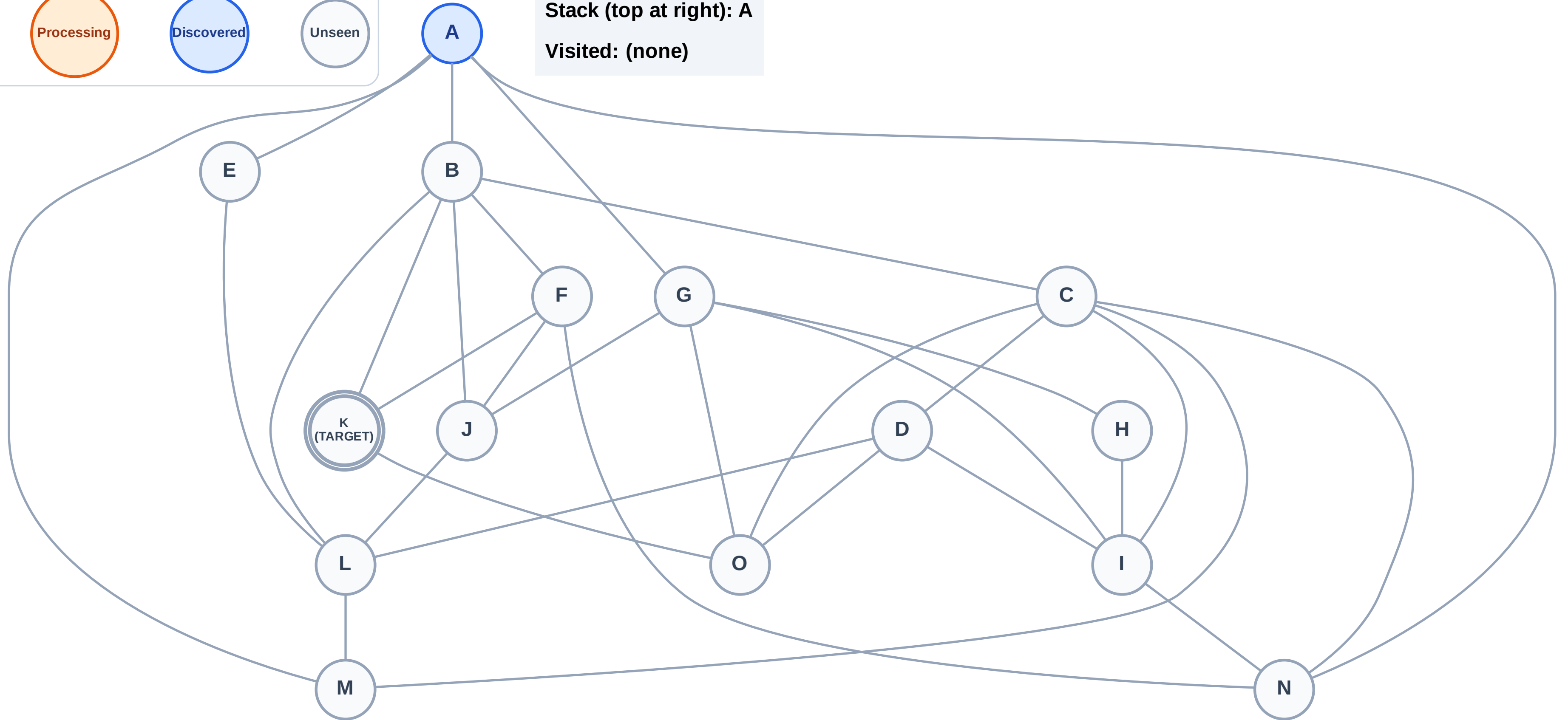
Discovered



Unseen

Stack (top at right): A

Visited: (none)



# DFS Traversal

Step 2: Process node A

Legend

Complete

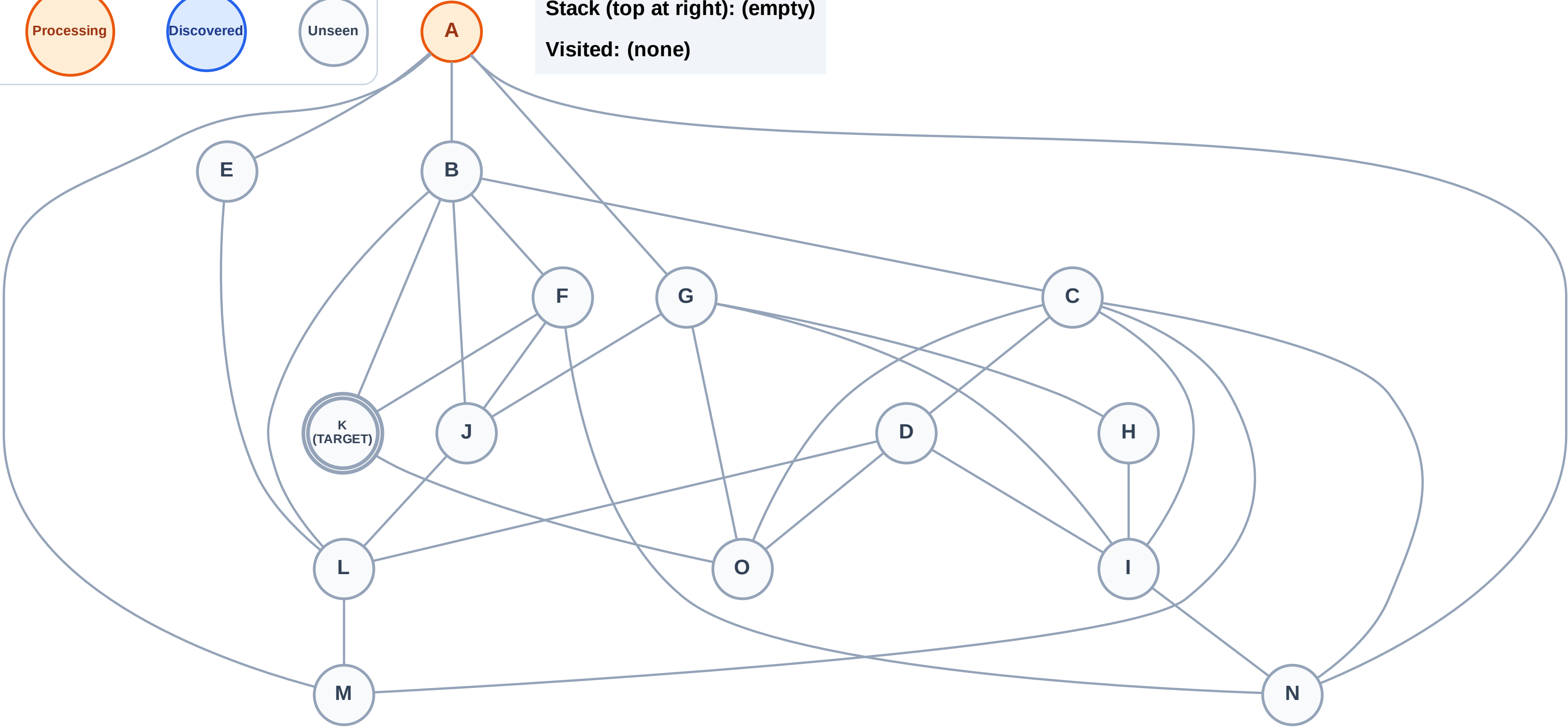
Processing

Discovered

Unseen

Stack (top at right): (empty)

Visited: (none)



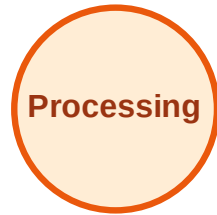
# DFS Traversal

Step 3: Discover N, M, G, E, B from A

## Legend



Complete



Processing



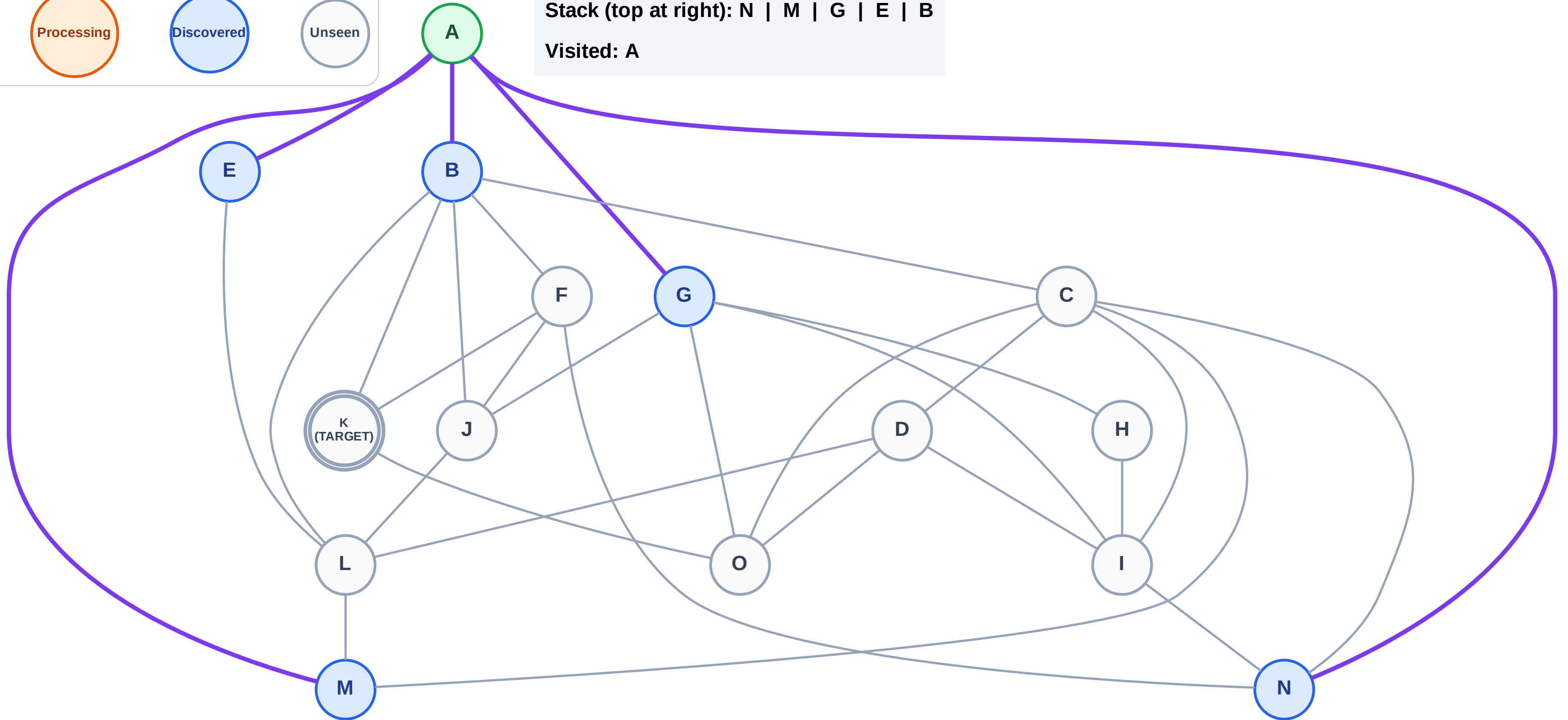
Discovered



Unseen

Stack (top at right): N | M | G | E | B

Visited: A



# DFS Traversal

Step 4: Process node B

Legend

Complete

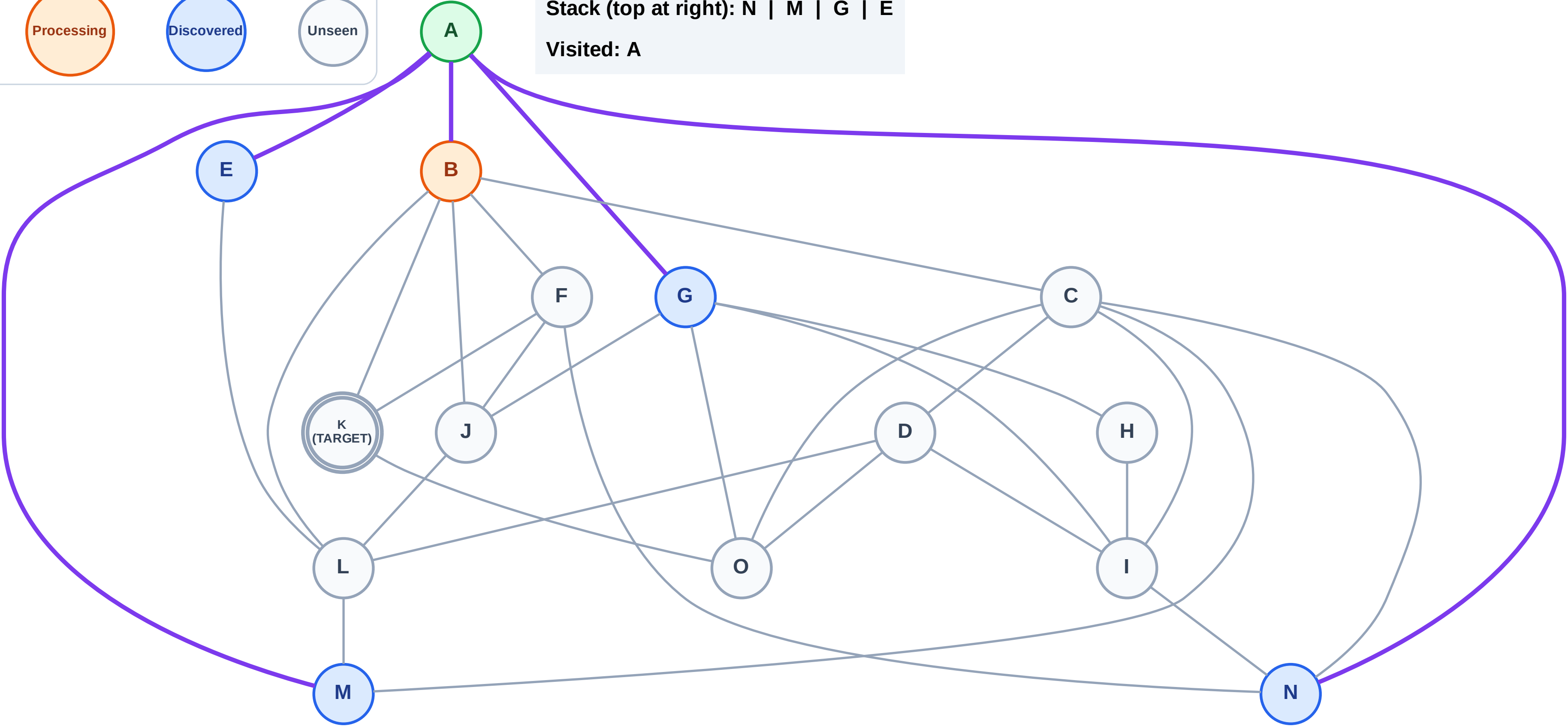
Processing

Discovered

Unseen

Stack (top at right): N | M | G | E

Visited: A



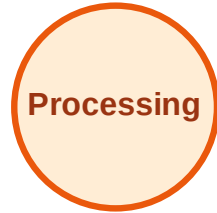
# DFS Traversal

Step 5: Discover L, K, J, F, C from B

## Legend



Complete



Processing



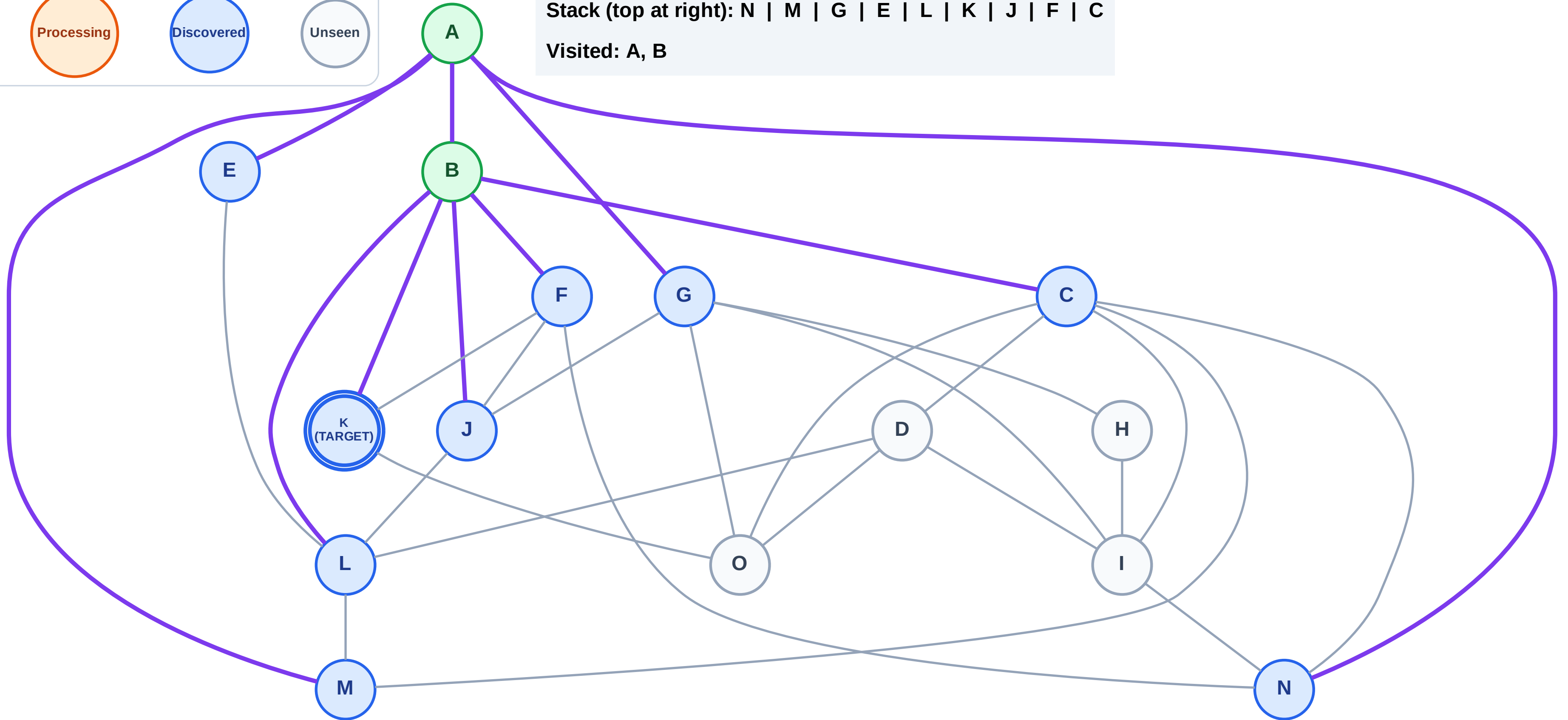
Discovered



Unseen

Stack (top at right): N | M | G | E | L | K | J | F | C

Visited: A, B



# DFS Traversal

Step 6: Process node C

Legend

Complete

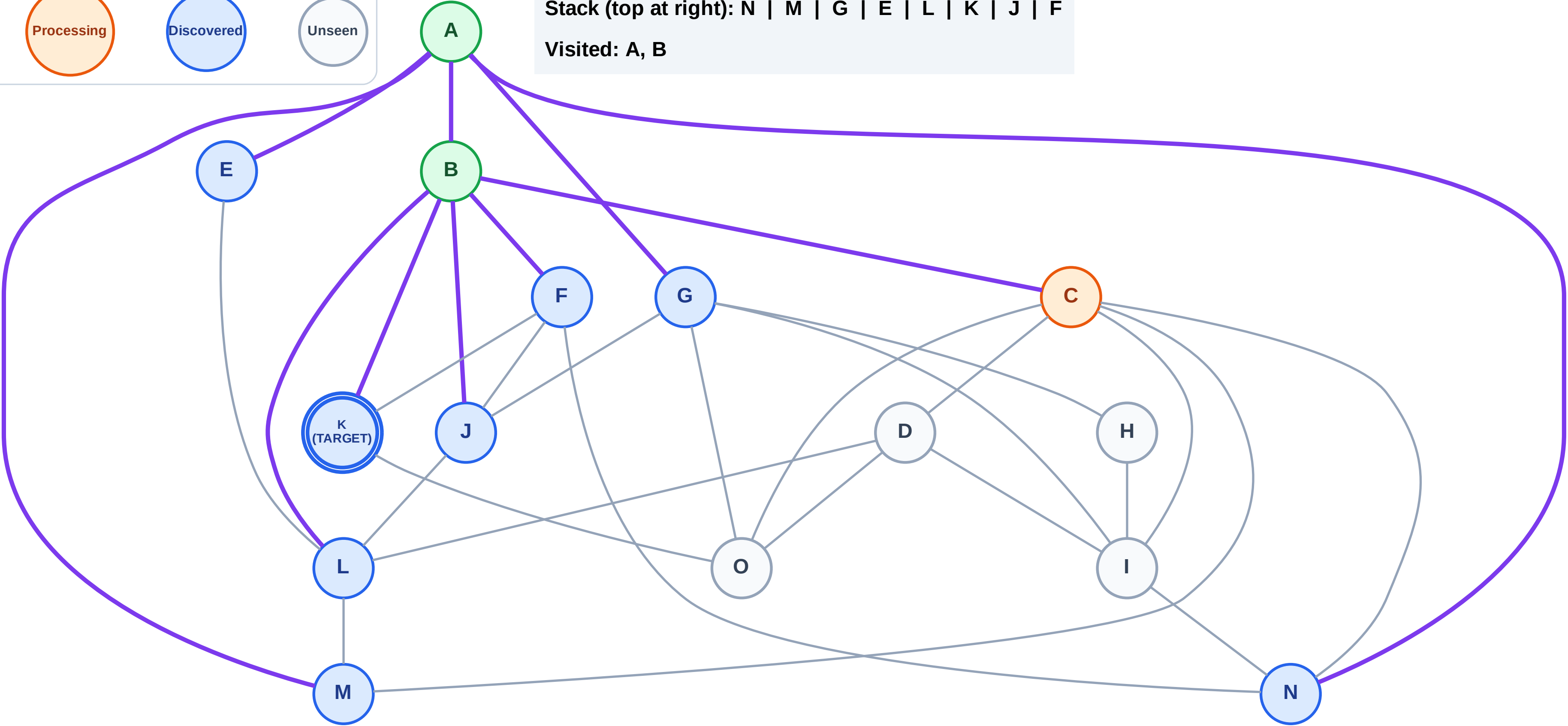
Processing

Discovered

Unseen

Stack (top at right): N | M | G | E | L | K | J | F

Visited: A, B



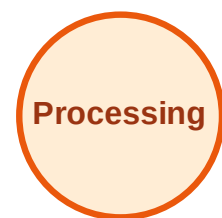
# DFS Traversal

Step 7: Discover O, I, D from C

## Legend



Complete



Processing



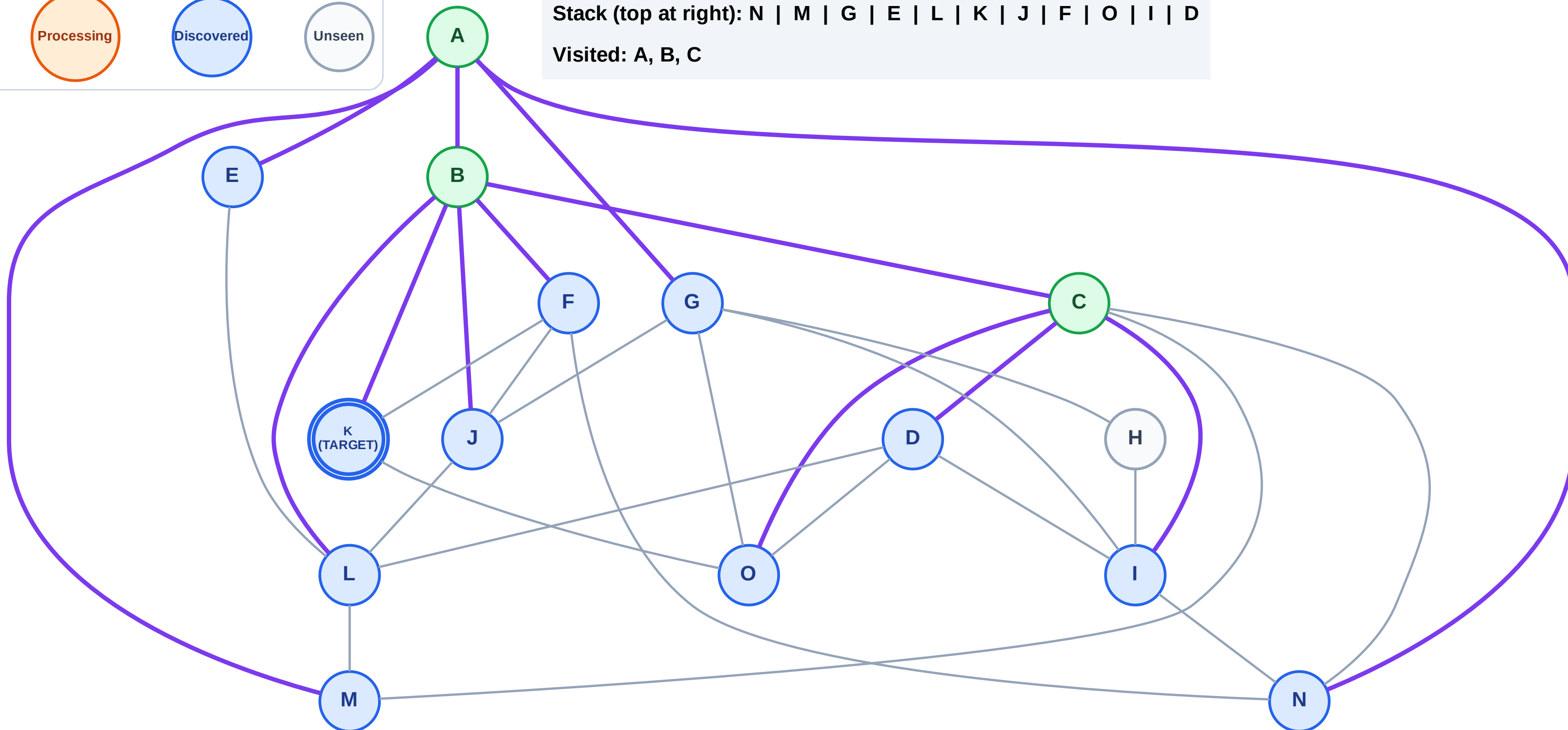
Discovered



Unseen

Stack (top at right): N | M | G | E | L | K | J | F | O | I | D

Visited: A, B, C



# DFS Traversal

Step 8: Process node D

Legend

Complete

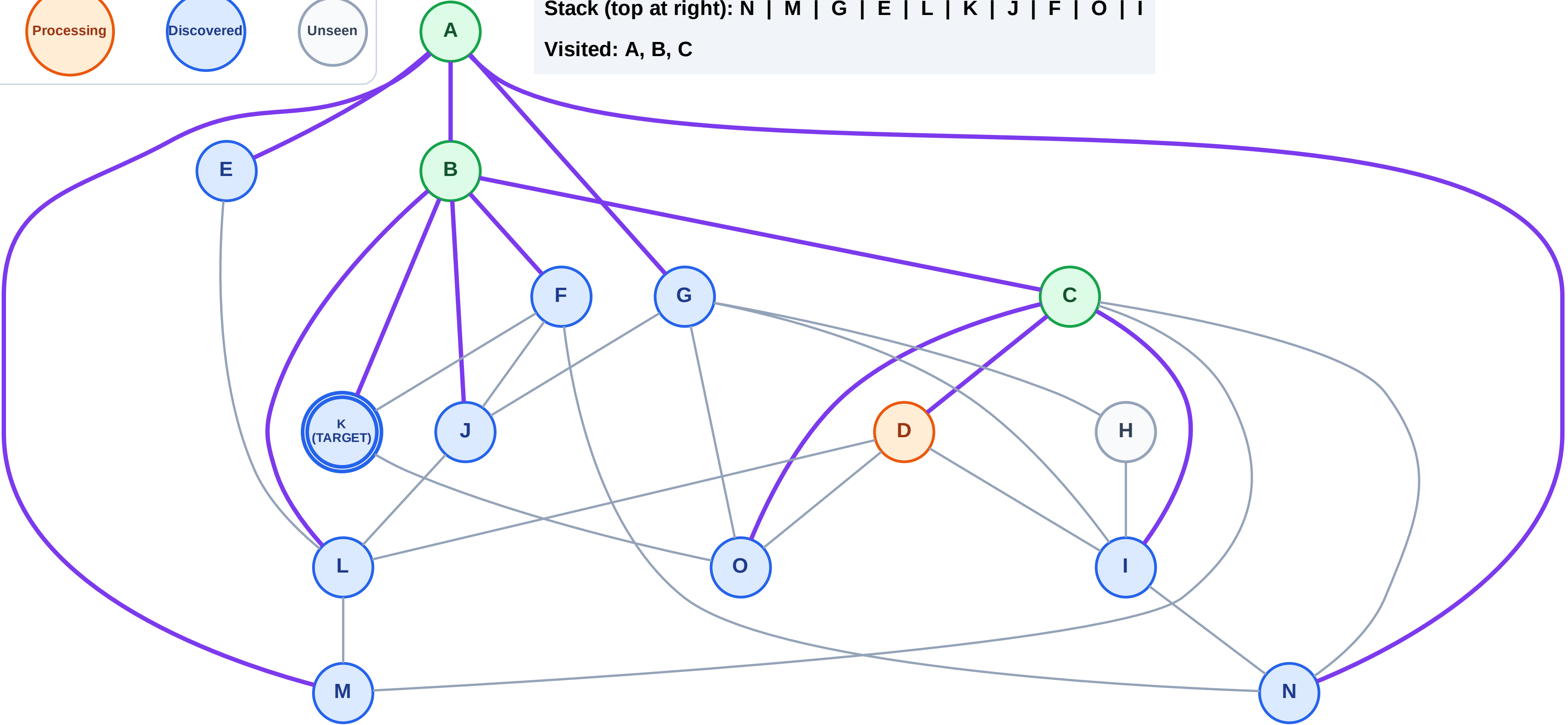
Processing

Discovered

Unseen

Stack (top at right): N | M | G | E | L | K | J | F | O | I

Visited: A, B, C



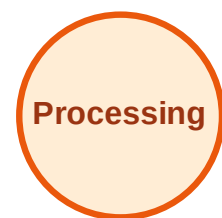
# DFS Traversal

Step 9: No new neighbors from D

## Legend



Complete



Processing



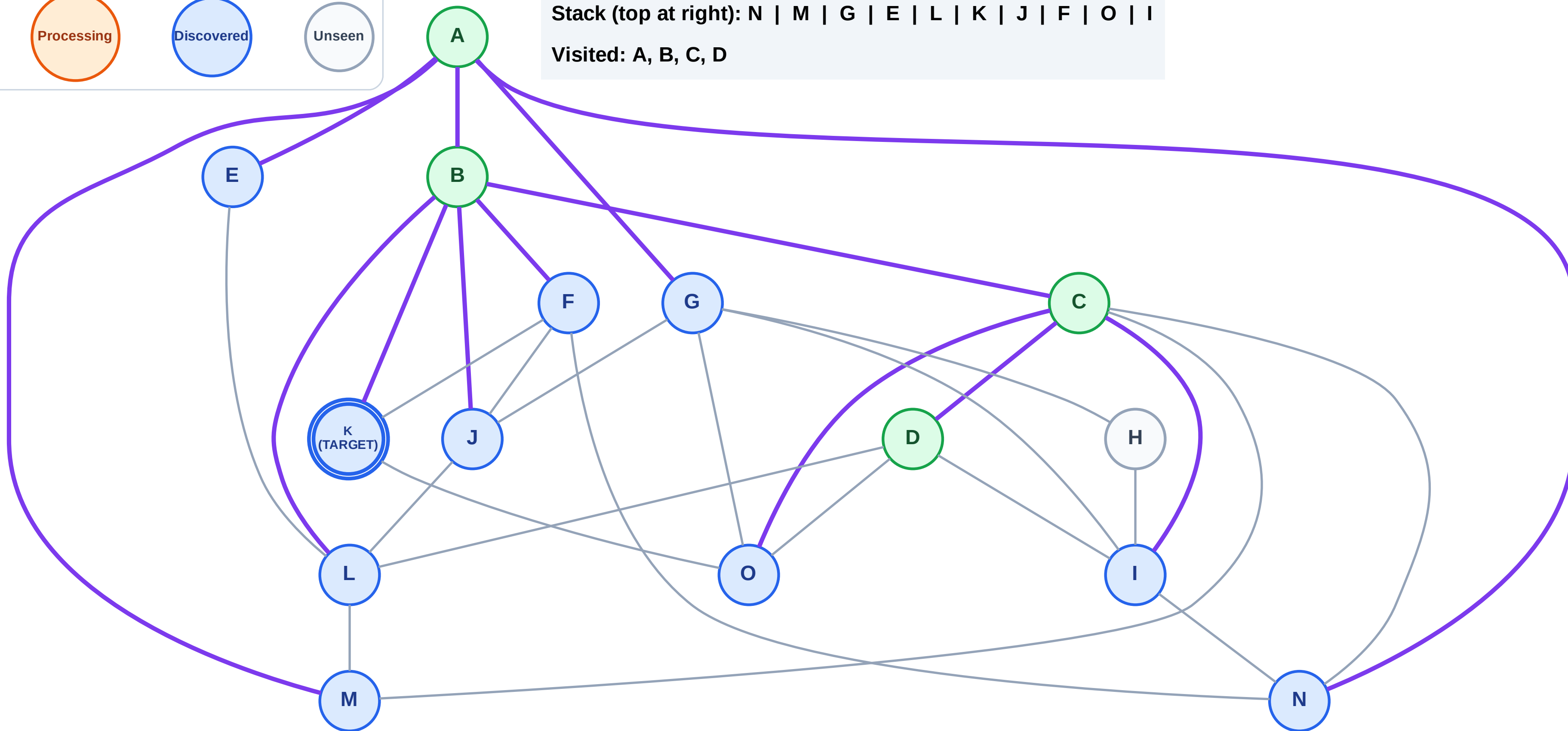
Discovered



Unseen

Stack (top at right): N | M | G | E | L | K | J | F | O | I

Visited: A, B, C, D



# DFS Traversal

Step 10: Process node I

Legend

Complete

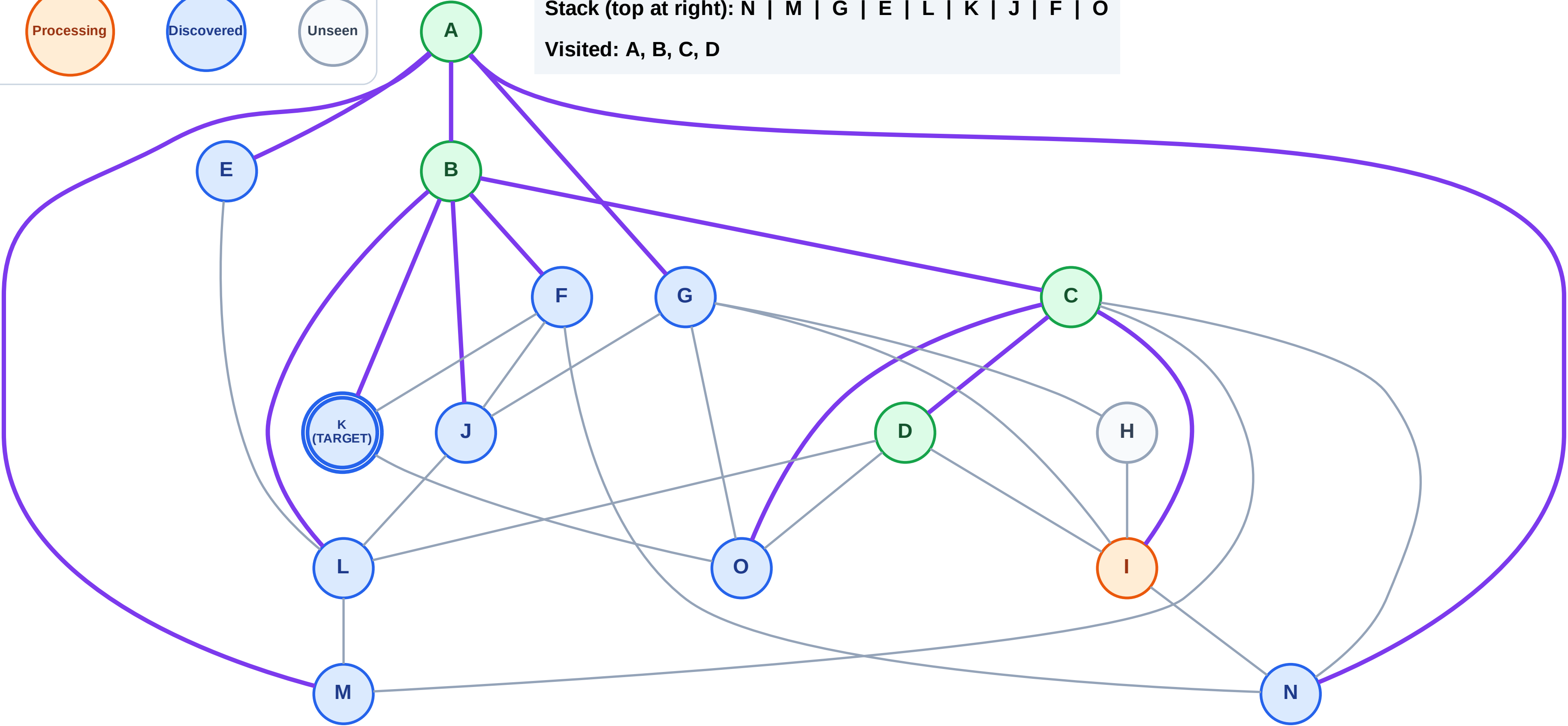
Processing

Discovered

Unseen

Stack (top at right): N | M | G | E | L | K | J | F | O

Visited: A, B, C, D



# DFS Traversal

Step 11: Discover H from I

Legend

Complete

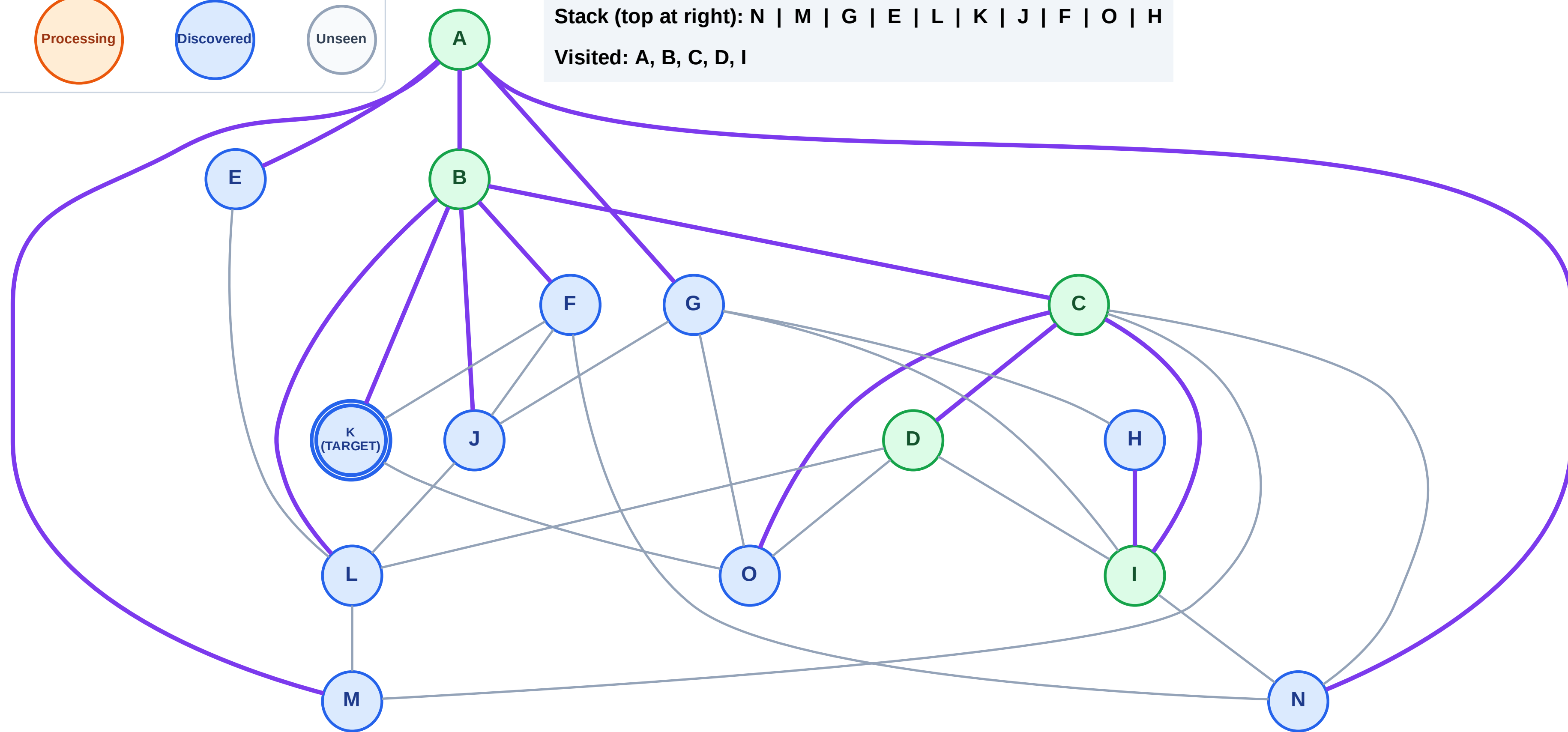
Processing

Discovered

Unseen

Stack (top at right): N | M | G | E | L | K | J | F | O | H

Visited: A, B, C, D, I



# DFS Traversal

Step 12: Process node H

Legend

Complete

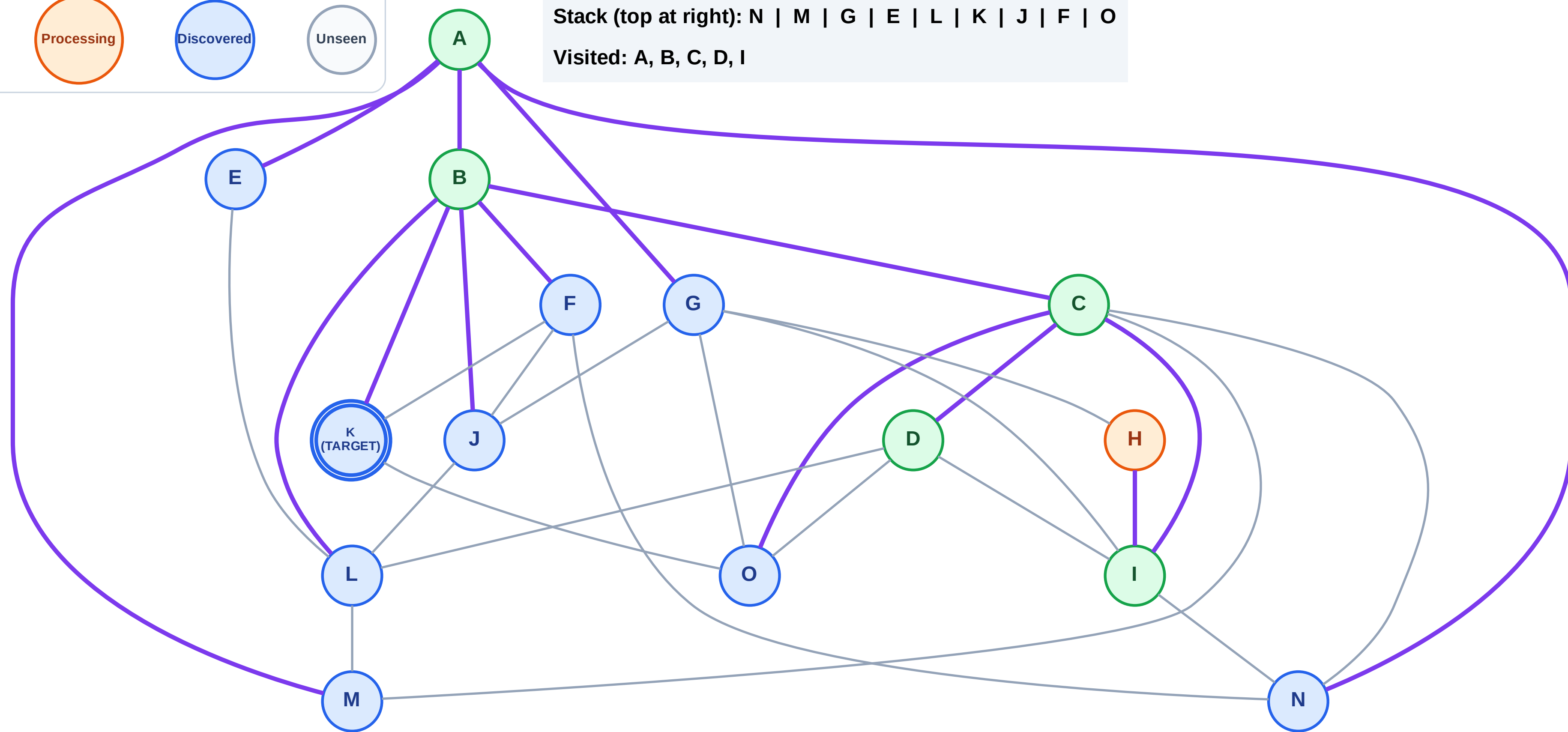
Processing

Discovered

Unseen

Stack (top at right): N | M | G | E | L | K | J | F | O

Visited: A, B, C, D, I



# DFS Traversal

Step 13: No new neighbors from H

## Legend



Complete



Processing



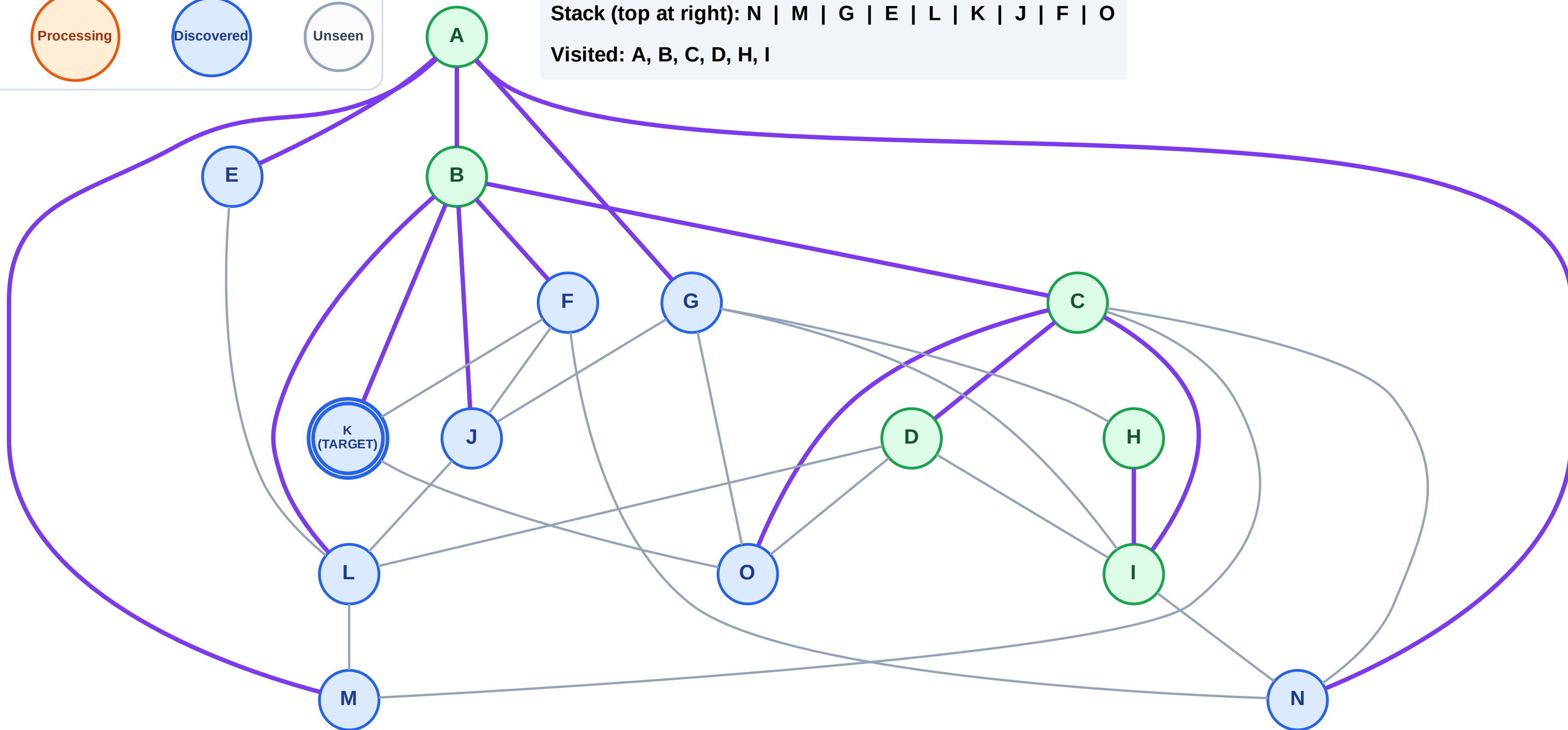
Discovered



Unseen

Stack (top at right): N | M | G | E | L | K | J | F | O

Visited: A, B, C, D, H, I



# DFS Traversal

Step 14: Process node O

Legend

Complete

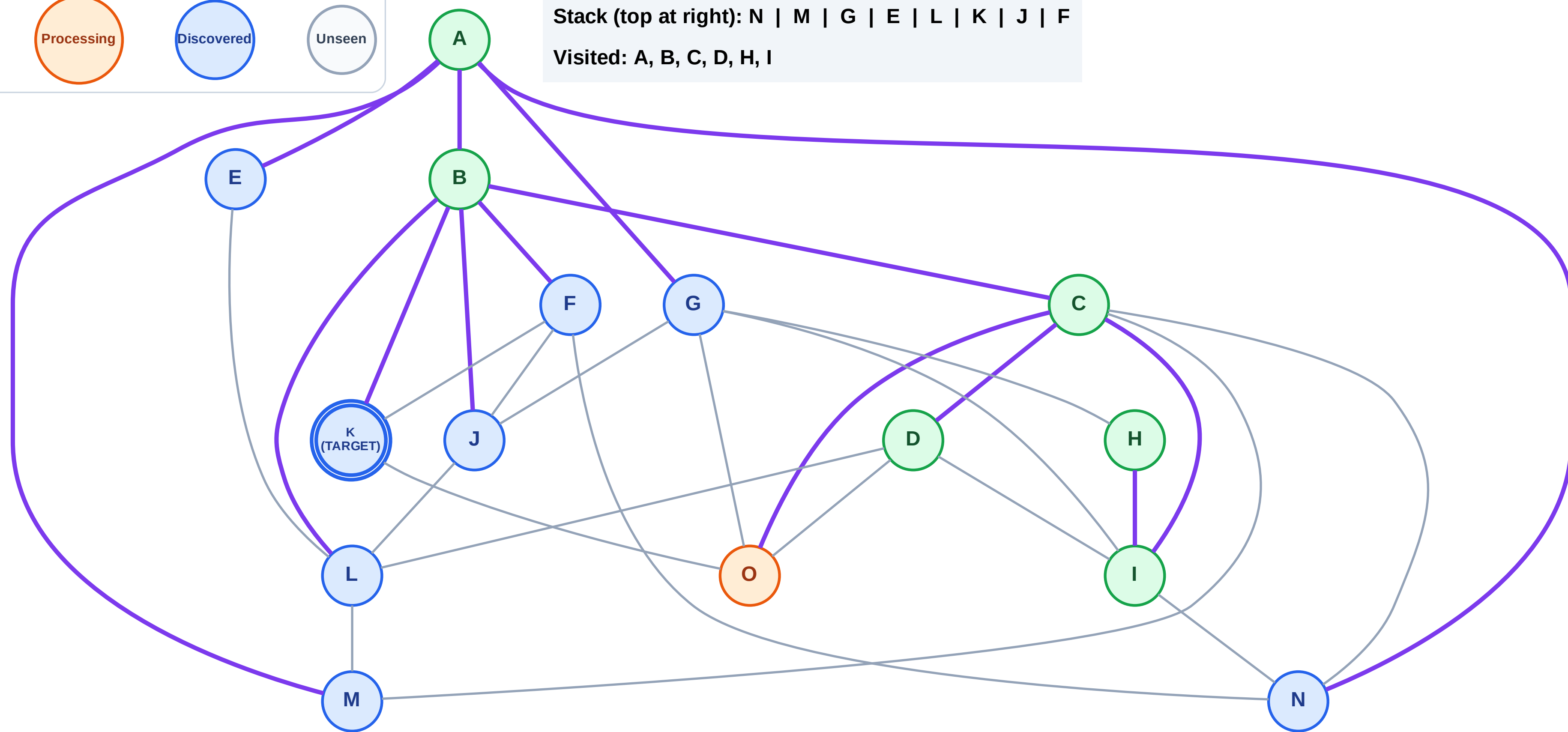
Processing

Discovered

Unseen

Stack (top at right): N | M | G | E | L | K | J | F

Visited: A, B, C, D, H, I



# DFS Traversal

Step 15: No new neighbors from O

## Legend



Complete



Processing



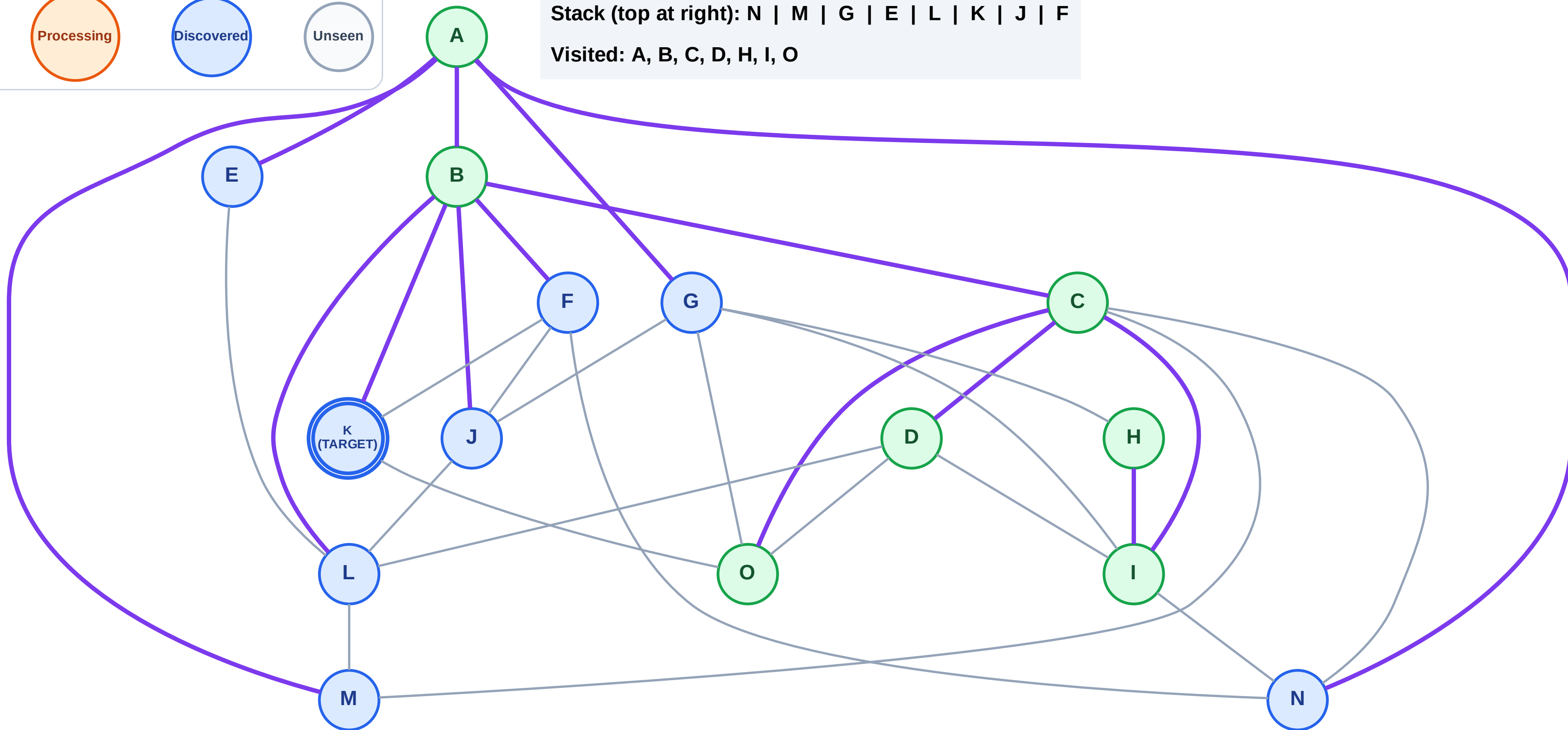
Discovered



Unseen

Stack (top at right): N | M | G | E | L | K | J | F

Visited: A, B, C, D, H, I, O



# DFS Traversal

Step 16: Process node F

Legend

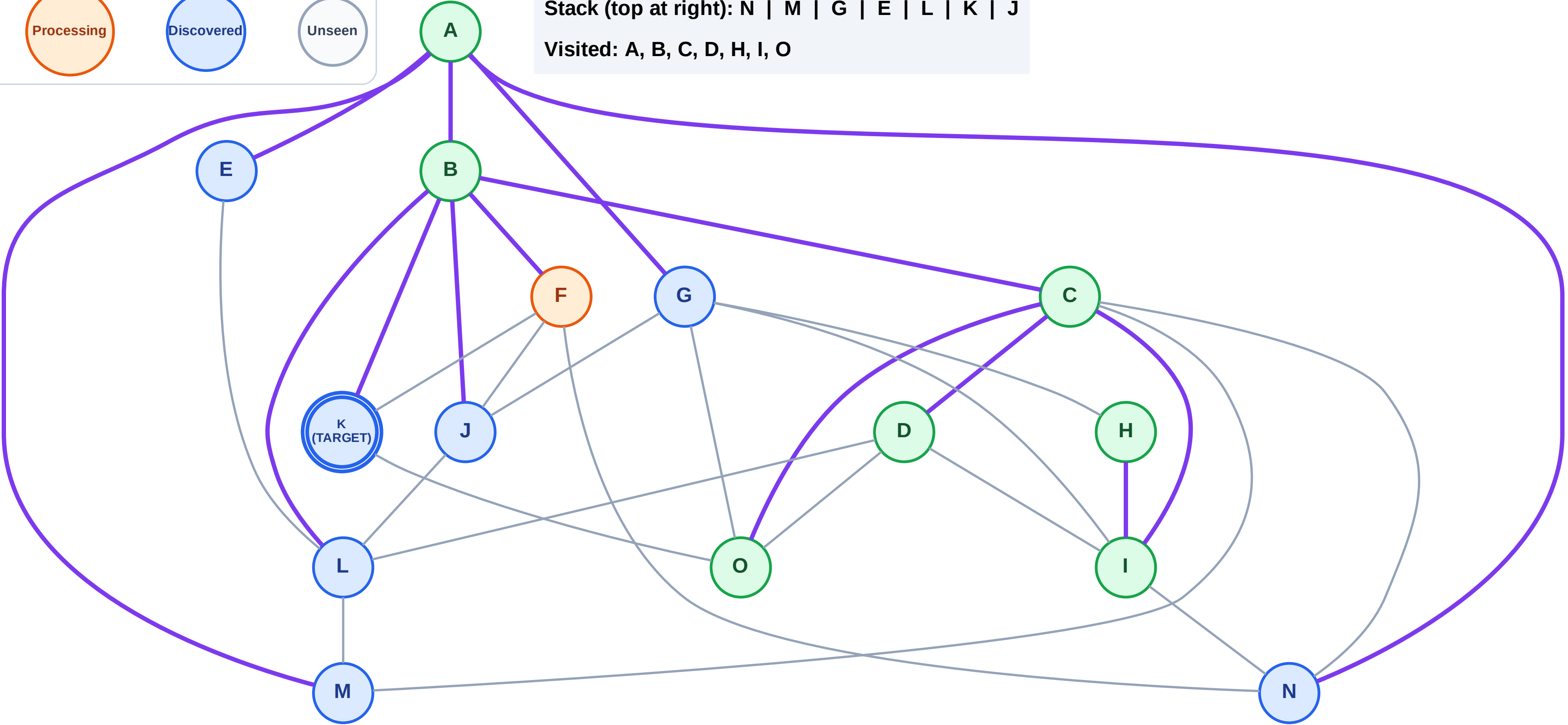
Complete

Processing

Discovered

Unseen

Stack (top at right): N | M | G | E | L | K | J  
Visited: A, B, C, D, H, I, O



# DFS Traversal

Step 17: No new neighbors from F

## Legend



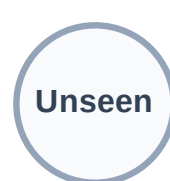
Complete



Processing



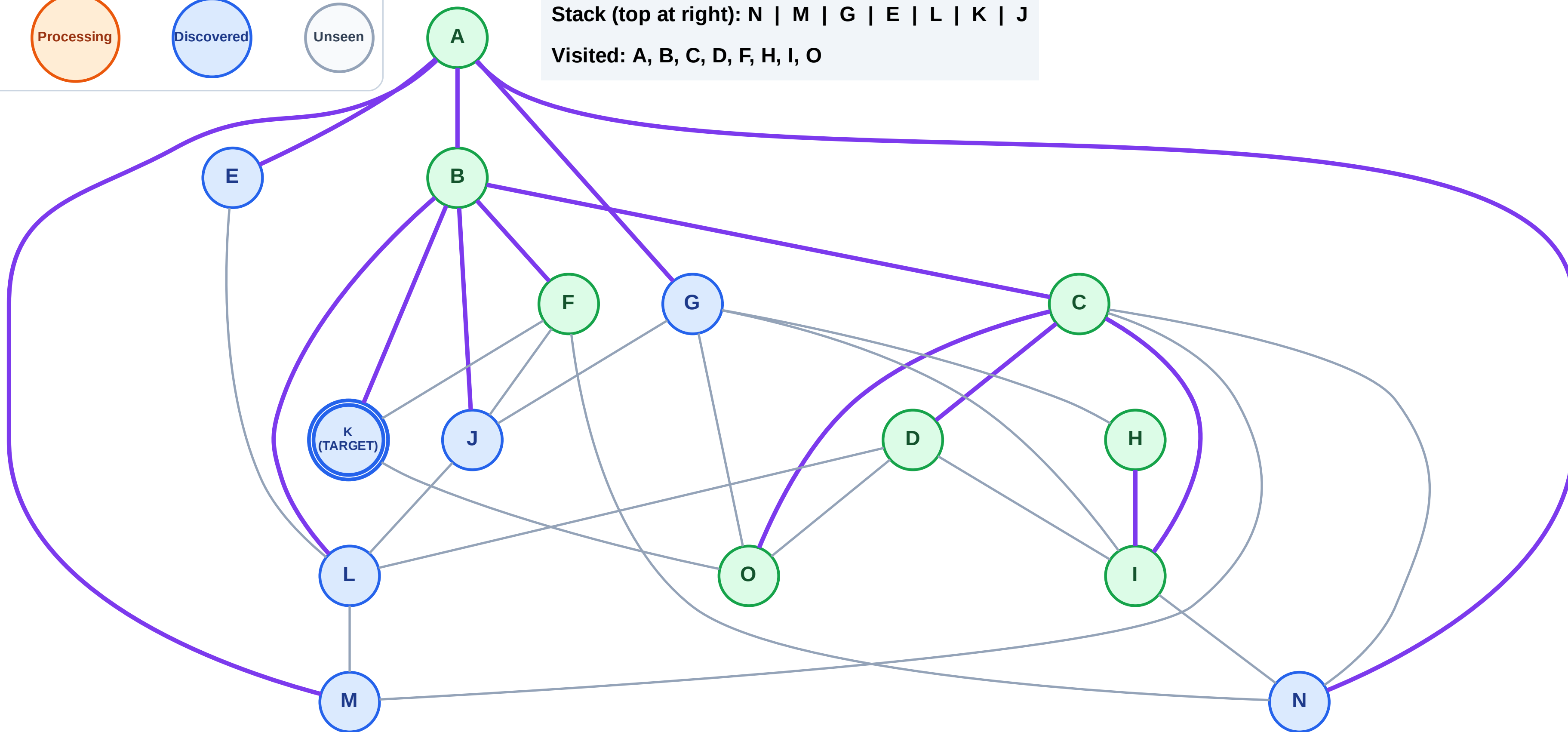
Discovered



Unseen

Stack (top at right): N | M | G | E | L | K | J

Visited: A, B, C, D, F, H, I, O



# DFS Traversal

Step 18: Process node J

Legend

Complete

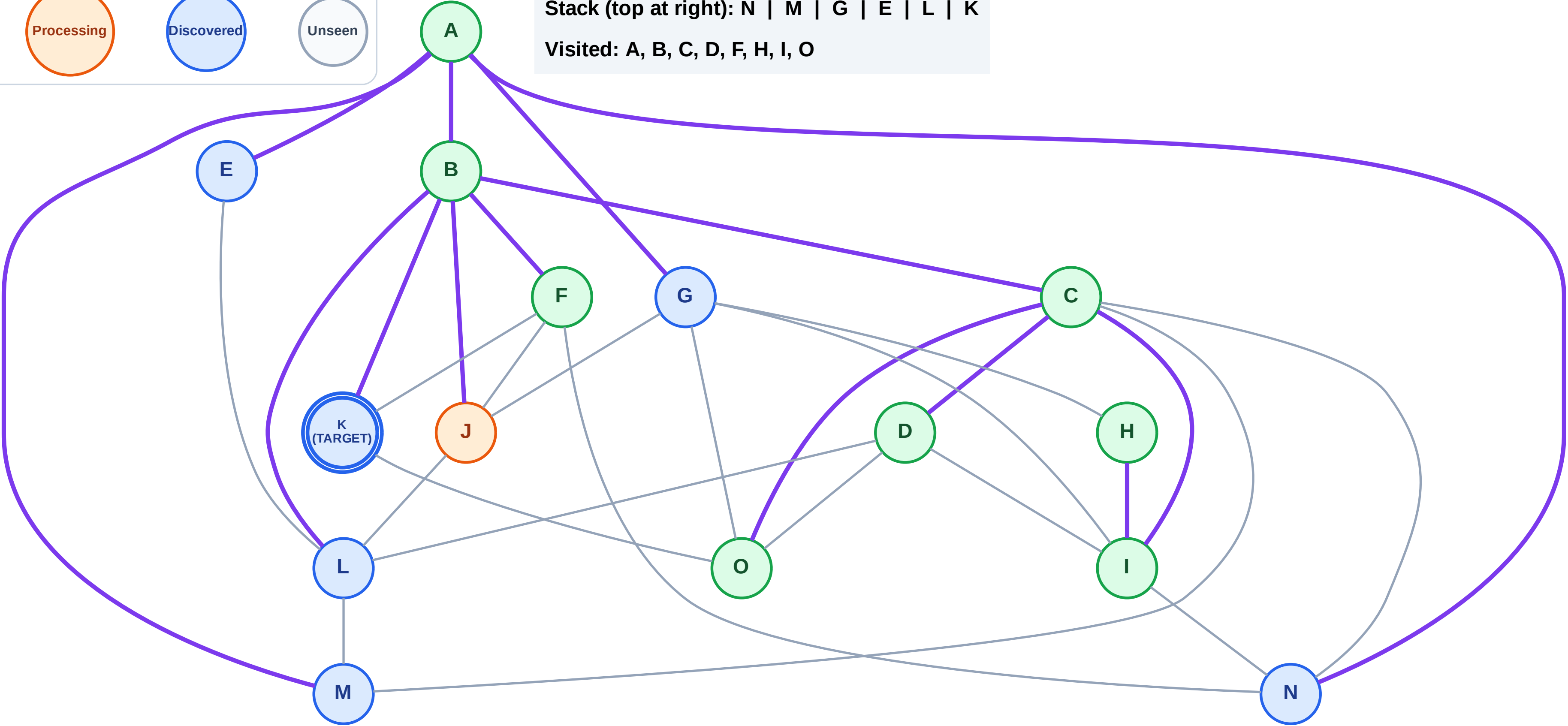
Processing

Discovered

Unseen

Stack (top at right): N | M | G | E | L | K

Visited: A, B, C, D, F, H, I, O



# DFS Traversal

Step 19: No new neighbors from J

## Legend



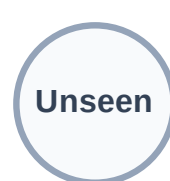
Complete



Processing



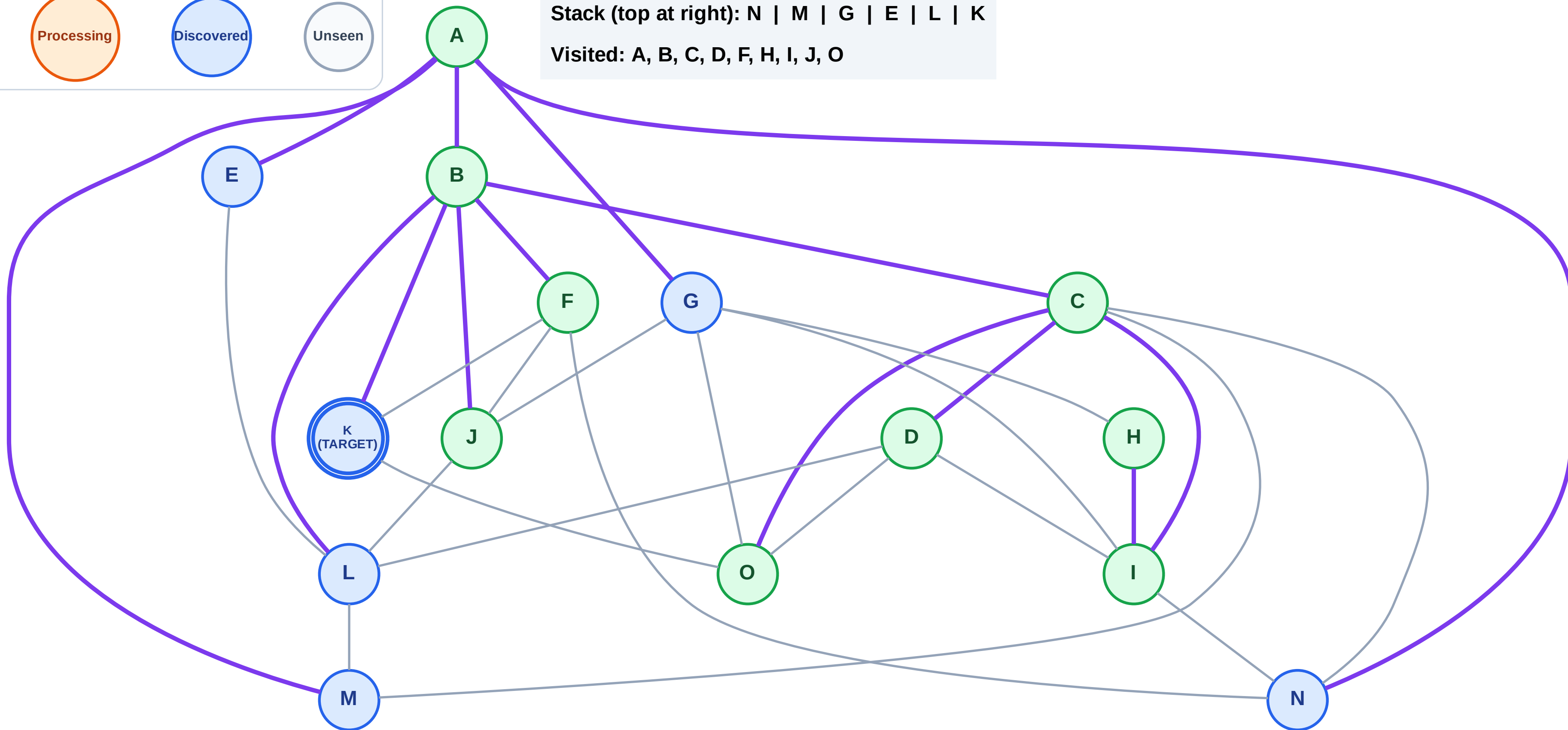
Discovered



Unseen

Stack (top at right): N | M | G | E | L | K

Visited: A, B, C, D, F, H, I, J, O



# DFS Traversal

Step 20: Process node K

Legend

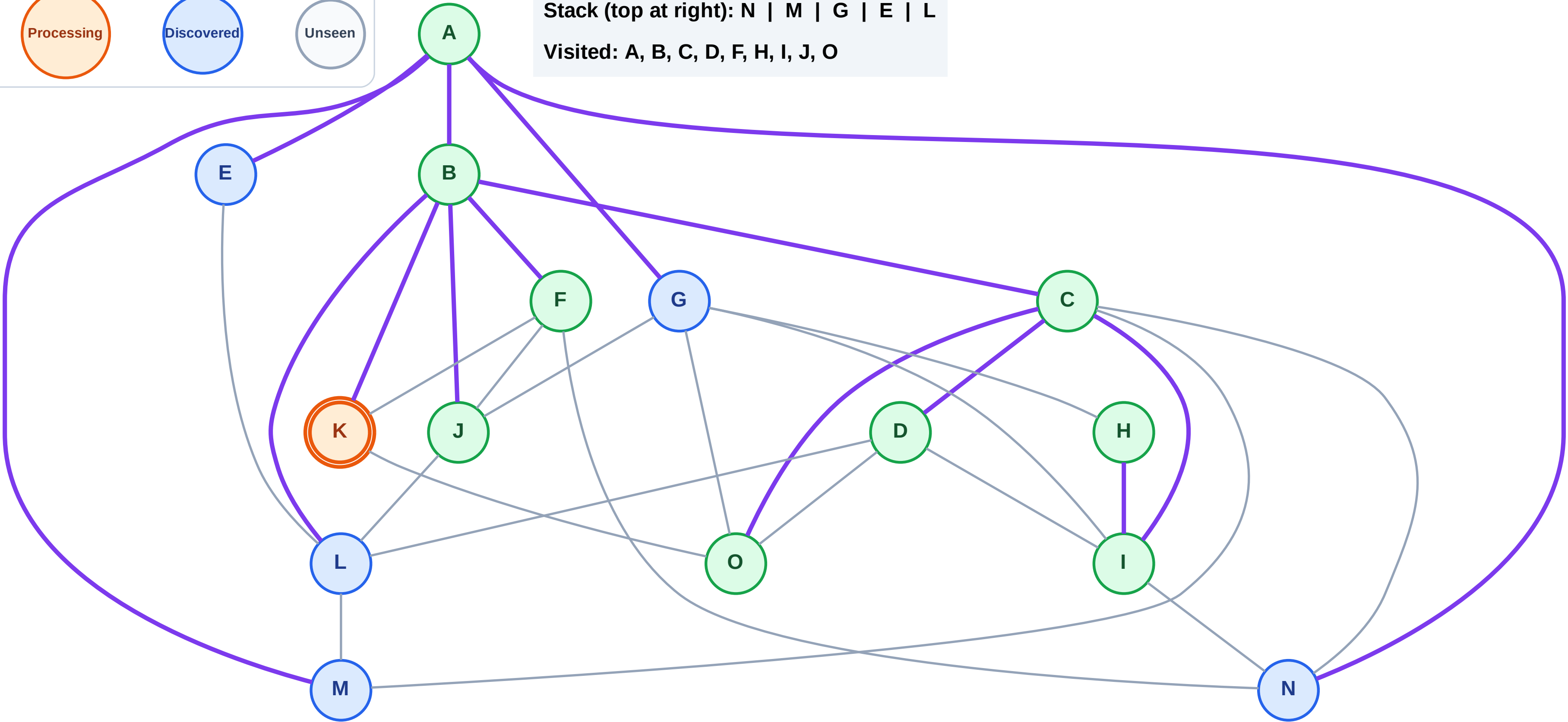
Complete

Processing

Discovered

Unseen

Stack (top at right): N | M | G | E | L  
Visited: A, B, C, D, F, H, I, J, O



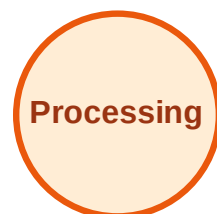
# DFS Traversal

Step 21: TARGET NODE K REACHED! Path (2 hops): A → B → K

## Legend



Complete



Processing



Discovered



Unseen

Stack (top at right): N | M | G | E | L

Visited: A, B, C, D, F, H, I, J, K, O

